Environment, Development and Sustainability
https://doi.org/10.1007/s10668-024-05549-7

An optimization approach for resilient and sustainable planning of intermodal transport

Asefeh Hasani Goodarzi[1] · Armin Jabbarzadeh[1] · Marc Paquet[1] · Michael Haughton[2]

Received: 5 July 2023 / Accepted: 9 October 2024
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract
Despite heightened concerns about freight network disruptions and their environmental impact, a critical knowledge gap persists in designing and operating resilient intermodal transport networks that simultaneously achieve two objectives: (a) economic performance and (b) low carbon emissions. This paper addresses this gap by introducing a novel bi-objective two-stage stochastic formulation for the network design and planning problem under disruption risks. In addition to covering both economic and emissions objectives, the model specifies the best set of actions to prepare for and respond to disruptions. These actions include the optimal number of vehicles from each transportation mode, the flow of containers on the transportation links, and the best route to deliver orders. To generate the non-dominated solutions, an augmented ε-constraint method is used on the data from a case study of the UK transportation network. The proposed approach contributes to the existing literature by focusing on the design and planning of intermodal networks that are not only resilient to disruptions, but are also cost-efficient, sustainable, and responsive. Numerical results reveal the potential of the proposed approach to significantly decrease carbon emissions by 62% with only a marginal increase in costs. The analysis of numerical results also shows a 16% cost savings resulting from strategic investments in appropriate preparedness and recovery actions. Additionally, the study indicates that despite a road mode capacity utilization of 98% to meet demands in disruption scenarios, it serves less than 9% of demand to align with environmentally sustainable objective.

Keywords Intermodal transportation · Resilience · Sustainability · Disruption risk · Mathematical programming

✉ Armin Jabbarzadeh
Armin.Jabbarzadeh@etsmtl.ca

Asefeh Hasani Goodarzi
asefeh.hassani-goodarzi.1@ens.etsmtl.ca

Marc Paquet
marc.paquet@etsmtl.ca

Michael Haughton
mhaughton@wlu.ca

[1] Department of Systems Engineering, École de Technologie Supérieure (ÉTS), University of Quebec, Montreal, Canada

[2] Lazaridis School of Business and Economics, Wilfrid Laurier University, Waterloo, Canada

Published online: 12 March 2025

1 Introduction

Freight transportation depends heavily on fossil fuel, so its continued growth translates to increased greenhouse gas (GHG) emissions (Chen & Zhang, 2023). As one of the largest GHG-emitting sectors, transport accounts for nearly 24% of global emissions, with road transport contributing around 75% (IEA, 2020). Despite a 5.8% reduction in carbon dioxide (CO_2) emissions in 2020 due to a 4% drop in energy demand during the COVID-19 pandemic, global CO_2 emissions from the transport sector increased by 8% in 2021 as pandemic restrictions were lifted (IEA, 2021). This trend continued into 2022, nearly restoring emissions to their 2019 levels (IEA, 2023).

The significant share of emissions and costs of transportation necessitates more efficient solutions and has prompted government policies (e.g., taxes) to foster environmentally responsible actions by companies (Fallahi et al., 2023). Such actions include transferring freight flows from road to lower emission modes such as rail or waterway (Sikder et al., 2022). Statistics from 2021 indicate that over 46% of total freight transportation in European Union countries occurred via road (Eurostat, 2023), with similar figures for the UK, where road transport accounted for around 91% of total transport emissions in 2019 (United Kingdom Government, 2022). Finding the right mix of transport modes using multimodal transport (i.e., the transfer of freight among at least two transportation modes on its origin-to-destination journey) is among solutions for reducing environmental impacts (Kuzmicz & Pesch, 2019). One example of multimodal transport is *intermodal* transport: this applies to situations in which operations at the freight transfer points (called intermodal terminals (Ishfaq, 2017)) involve keeping freight in the same loading unit (e.g., a container) throughout the journey (Mostert et al., 2017).

This is beneficial as it ensures cargo safety from loss and damage, reduced cargo handling costs, and efficient fuel consumption in both domestic and inter-continental networks (He et al., 2015). Yet, as the following examples show, transportation networks remain vulnerable to disruptions, which can cause efficiency-reducing cancellation or revision of freight distribution plans:

- 3-week Montreal rail blockade shrank Canada's 2020 first quarter GDP growth by 0.3% (Ke, 2020).
- COVID-19 disrupted the supply chains of 94% of USA Fortune1000 companies; 75% of companies adversely affected; 55% lowered (or plan to lower) their growth outlooks (Accenture report, 2021).
- 6-day Suez Canal blockage in 2021 prevented roughly US$9.6 billion in trade (BBC News, 2021; The Guardian, 2021).

Integrated approaches that concurrently address both disruptions and environmental sustainability issues are insufficiently explored in the existing freight network literature. This paper seeks to bridge this gap by focusing on the design and planning of multi-modal networks that not only demonstrate resilience to disruptions but are also cost-efficient, sustainable, and responsive. Specifically, we propose a two-stage stochastic model with two objectives: (i) minimizing the expected system (network) cost under different disruption scenarios and (ii) minimize the expected CO_2 equivalent emission (CO_2e). In addition to identifying optimal proactive and reactive strategies for coping with disruptions, the proposed model determines the optimal number of vehicles from each transportation mode, the flow of containers on transportation links, and the optimal routes for order delivery.

The model provides insights into key interactions, such as those between environmental sustainability and resilience, and enables us to answer the central research question: "*How do (a) disruption uncertainties affect system cost, resilience, and environmental sustainability, and (b) sustainability measures impact cost-efficiency?"*.

Therefore, the proposed research contributes to the literature in the following ways. Firstly, unlike the existing literature on intermodal transportation that often overlooks the joint incorporation of sustainability and resilience in planning, this paper integrates sustainability and resilience measures into the planning model. By accounting for random disruptions and carbon emissions, the proposed approach determines optimal proactive and reactive actions to enhance the system's resilience and explores their impact on sustainability performance. Secondly, compared to prior studies, the proposed approach provides a more comprehensive and precise economic analysis for intermodal transport. This is achieved by considering various cost components, including the fixed and variable cost of shipping commodities, the cost of implementing proactive and reactive actions, and transfer costs for commodities in terminals. The proposed approach allows for trade-offs between economic and environmental sustainability objectives for companies facing budgetary constraints. Thirdly, the proposed approach can incorporate different responsiveness and service level measures, such as lost sales and service time, in the analysis. Examining the UK transportation network as a case study, our investigation explores how the proposed approach supports transport companies in decision-making, aiming to minimize total cost and emissions, address disruption risks, and achieve desired service levels.

Our study to address these research contributions is reported in the remainder of this paper as follows: Sect. 2 (literature review); Sect. 3 (the problem statement and methodology); Sect. 4 (a case study from the UK transportation network, along with computational experiments and analysis); Sect. 5 (discussion and managerial implications); and Sect. 6 (conclusions).

2 Literature review

There are several recent comprehensive literature reviews on multimodal freight transportation planning; e.g. Caris et al. (2008), SteadieSeifi et al. (2014), Baykasoğlu et al. (2018) and Archetti et al. (2022). Crainic and Kim (2007) also published a book chapter detailing basic concepts of intermodal transportation with a focus on operations research models and methods.

The problem of interest in the present study lies at the tactical planning level, focusing on optimal infrastructure usage decisions: i.e., selecting services and associated transportation modes, allocating modes to orders, and planning their itineraries and frequency. We now present a two-part literature review to address the two research streams of most relevance to our study: disruption management and environmental sustainability. Following this, we perform a comparative analysis of the literature, highlighting research gaps and elucidating the contributions of our work.

2.1 Disruption management in multimodal transportation literature

Table 1 presents a concise overview of pertinent publications in the disruption management of multimodal systems. Following this, we will provide brief reviews of selected publications and proceed with a comparative analysis within this field in Sect. 2.3.

Table 1 The review of related papers in disruption management of multimodal systems

	Field/decision making problem	Disruptions	Performance measure(s)	Modelling approach	Solution method
Ishfaq (2012)	• Resilience • Flexibility • Link assignment (shortest path)	Disruptions at links	Transportation cost (including transfer cost)	Two models to provide an alternate route in disruption case	Coded in CPLEX
Chen and Miller-Hooks (2012)	• Flow network planning • Resilience • Budget constraints • Recovery actions	• Disruptions at links by: • Bombing • Terrorist attack • Flooding • Earthquake • Intermodal terminal attack	Resilience level	Stochastic mixed-integer program	A technique based on Benders' decomposition, column generation, and Monte Carlo simulation
Miller-Hooks et al. (2012)	• Flow network planning • Resilience • Budget constraints • Recovery actions • Preparedness actions	• Disruptions at links by: • Bombing • Terrorist attack • Flooding • Earthquake • Intermodal terminal attack	Resilience level	Two-stage stochastic program	Benders' decomposition and Monte Carlo simulation
Uddin and Huynh (2016)	• Route planning	• Disruptions at: • links • Intermodal terminal	Transportation cost (including transfer cost and penalty cost)	Stochastic mixed-integer programming	Sample Average Approximation (SAA) algorithm
Azad et al. (2016)	• Itinerary planning • The optimal number of trains in train services	• Disruptions in service legs and train services	Transportation cost	Optimization-based predictive method to make a trade-off between mitigation cost and the expected disruption cost	Coded in MATLAB, and CPLEX 12.6.0

Table 1 (continued)

	Field/decision making problem	Disruptions	Performance measure(s)	Modelling approach	Solution method
Fotuhi and Huynh (2016)	• Location of new terminals • Network expansion decisions • Critical rail links to retrofit	• Demand uncertainty • Disruption at terminals • Disruption at links	Establishing cost of new terminals and expanding the existing terminals and retrofit cost	Robust method (min–max regret)	Hybrid GA. Column generation
Fotuhi and Huynh (2018)	• Location of terminals in expansion period • Expansion design • Link retrofitting • Route planning	• Uncertain demand • Disruption at terminals • Disruption at links	• Expected routing cost and expansion cost • Penalty cost for unmet demand	Probabilistic robust optimization	Hybrid SA
Ishfaq (2017)	• Route planning	• Removing nodes in the disrupted region and all demand flows originating (ending) from (at) these cities • Removing arcs to/ from the identified cities	Transportation cost (including transfer cost)	Stochastic programming	Coded in CPLEX
Sarhadi et al. (2017)	• Protection planning • Terminal fortification	• Disruptions at terminals	• Transportation cost (including penalty costs for late deliveries) • Cost of fortifying terminals	Tri-level optimization model	A Benders' decomposition-based heuristic
Jabbarzadeh et al. (2020)	• Hazmat network • Route planning	• Disruptions at service legs (and their capacity)	• Transportation cost (including cost of fulfilling demand through a third party and adding extra capacities) • Hazmat risk	Two-stage stochastic programming	Augmented ε-constraint approach

Table 1 (continued)

	Field/decision making problem	Disruptions	Performance measure(s)	Modelling approach	Solution method
Uddin and Huynh (2019)	• Route planning • Reliable routing	• Disruptions at: • links, • Intermodal terminals (effect on capacity and travel/transfer time)	Transportation cost (including transfer cost and penalty cost for unmet demand)	Stochastic MIP (expected values for random variables)	Coded in Python and CPLEX
Maiyar and Thakkar (2019b)	• Hub location-allocation • Route planning • Sustainability	Disruptions at hubs	• Hub location cost • Transportation cost (including environmental and social costs)	Stochastic programming (in the case of disruption in a hub, demand is satisfied by rerouting through emergency hubs)	PSODE
Maiyar and Thakkar (2019c)	• Hub location-allocation • Route planning • Sustainability	Disruptions at hubs	• Hub location cost • Transportation cost (including environmental cost)	Robust optimization (min of total relative regret)	PSODE
Ghaderi and Burdett (2019)	• Hazmat network • Location-Routing	• Disruptions at transfer yards	• Transportation cost • Transportation risk	• Two-stage stochastic programming	• Likelihood Sampling (MLS) • SAA • Combined method of MLS and SAA
Ke (2020)	• Hazmat network	Disruptions at intermodal yards	• Transportation cost (including penalty cost for delay in delivery) • Risk level	Scenario-based robust optimization	Coded in Python 2.7
Rosyida et al. (2020)	• Route planning • VRPTW	• Disruptions at links • Disruptions at customers	• Transportation cost (including transfer cost and penalty cost of violating time window)	Two models including the original and the recovery model	GA, SA

Table 1 (continued)

	Field/decision making problem	Disruptions	Performance measure(s)	Modelling approach	Solution method
Wagenaar et al. (2021)	• Flow network planning • Customer service levels • Timetabled services	• Disruptions at links	• Costs of hiring additional trains • Trip cancellation costs • Tactical planning deviations from strategic commitments	Scenario-based modelling	A scenario-based heuristic
Ke and Verma (2021)	• Route planning • Recovery planning	• Disruptions at intermodal terminals	• Transportation cost (including fixed cost to provide different types of intermodal trains)	Two MIP models for normal and post-disruption operations	Coded in Python, solved with Gurobi
Li et al. (2023)	• Route planning	• Disruptions at storage	• Transportation cost (including inventory holding cost)	Scenario-based modelling	Coded in Python, solved with SCIP
Akyüz et al. (2023)	• Network flow planning • Route replanning	• Disruptions due to the arrival of new orders, fluctuations in shipment quantities, delays, and service cancellations	Transportation cost	Single objective model	Column generation
Nocera et al. (2024)	• Supply network planning	Disruption at the distribution logistics hubs	• Resilience level • Revenue • Transportation cost	Single objective model	Monte-Carlo simulation
Current study	• Route planning • Sustainability • Resilience • Recovery actions • Preparedness actions • Budget constraints • Responsiveness and service level	Disruptions at links by: • Bombing • Terrorist attack • Flooding	• Transportation cost (including transfer cost and penalty cost for unmet demand) • Resilience level • Cost of preparedness and recovery activities Emission value	Scenario-based two-stage stochastic programming	Augmented ε-constraint

The typical categories of disruption management strategies are: (i) *mitigation strategies* (which involve preparedness actions before a disruption) including increasing safety stock and/or capacity and multiple sourcing; (ii) *recovery strategies* (after a disruption occurs) such as using alternative shipment sources, routes and rescheduling in recovery periods, and (iii) *passive acceptance* (Hasani Goodarzi et al., 2020). The literature provides informative examples of other disruption management strategies: e.g., negotiate backorder arrangements with customers; obtaining needed resources from backup suppliers (Ahmadi-Javid & Seddighi, 2013); reroute shipments (Jabbarzadeh et al., 2020); add capacity at existing intermodal terminals (Ke, 2020); retrofit existing transportation links (Fotuhi & Huynh, 2018); add capacity on safer links (Jabbarzadeh et al., 2020); build temporary roadways (Chen & Miller-Hooks, 2012).

Chen and Miller-Hooks (2012) and Miller-Hooks et al. (2012) are among the first researchers to address disruptions in multimodal/intermodal transportation networks. Both studies utilized a simplifying assumption, restricting the impact of disruptions exclusively to rail links. Additionally, in their models, intermodal transfers were considered the sole mechanism for recovery in the face of disruptions. Uddin and Huynh (2016) formulated a stochastic mixed-integer program (MIP) to determine the routes of multi-commodity freights in a road-rail intermodal network subject to disruptions. Unlike previous works, the authors proposed a link-oriented formulation. Later, Uddin and Huynh (2019) investigated a reliable routing plan in a road-rail intermodal network with disruptions that degrade the capacity of the transportation modes and intermodal terminals. In the strategic level of decision making, Fotuhi and Huynh (2016) proposed a robust mixed-integer linear programming (MILP) for rail-road intermodal network expansion problems considering both link and terminal disruptions. They then extended their previous work, in Fotuhi and Huynh (2018), through a probabilistic robust model to address the optimization decisions, including the best locations of new intermodal terminals, the amount of extra capacity to add at existing terminals, and the existing rail links to retrofit.

Another robust approach was presented by He et al. (2021) to handle disruptions of infrastructure elements and capacity degradation of pathways in a rail-road-inland waterway network. Network robustness was evaluated based on the increase in total travel time resulting from perturbations. Hrušovský et al. (2021) introduced a novel real-time decision support system utilizing a hybrid simulation–optimization approach at the strategic level. This system integrated offline planning with online re-planning using real-time data on unforeseen events in the intermodal transportation network. When disruptions occurred, the system identified affected services and applied the most effective re-planning policy to mitigate their impact. In a similar vein, Akyüz et al. (2023) presented a replanning decision tool, as an alternative to the mathematical model proposed by Hrušovský et al. (2021), for managing disruptions. They employed the column generation method to enhance the efficiency of the replanning process. In an application-based study, Pudasaini (2021) conducted research on the planning of an uncertain downstream petroleum supply chain. The study developed a multi-objective stochastic model considering uncertain demand, emphasizing the strategic aspects of decision-making in the face of disruptions.

2.2 Sustainability in multimodal transportation literature

Table 2 provides a summary of the relevant literature on sustainability in multimodal transportation. Subsequently, we will conduct a brief review of selected publications and perform a comparative analysis in Sect. 2.3.

Table 2 A review on studies in sustainable multimodal transportation

Authors	Field/decision making problem	Disruption	CO_2/CO_2e emission optimization		Due date consideration	Performance measure(s)	Modelling approach	Solution method
			Bi-objective optimization	Charging method				
Qu et al. (2014)	• Route planning		✓	✓		Transportation cost (including transfer cost)	Non-linear integer programming	ε-constraint
Liotta et al. (2015)	• Integrating supply, production, transportation in strategic and tactical level			✓	✓	• Sourcing, production and transportation costs • Emission cost	Single objective	Coded in CPLEX
Demir et al. (2016)	• Route planning			✓	✓	• Time-related cost (charges for delayed deliveries) • Emission costs	Stochastic single objective	SAA method
Hrušovský et al. (2016)	• Route planning			✓	✓	• Time-related cost (in-transit inventory costs and charges for delayed deliveries) • Emission costs	Hybrid optimization and simulation approach	Simulation software
Mostert et al. (2017)	Location-allocation of terminals		✓			• Transportation cost (including transfer cost) • Emission value	Bi-objective model (total transportation costs and CO_2 emissions-related costs)	ε-constraint
• Demir et al. (2019)	• Route planning (at tactical and operational level)		✓			• Transportation cost (including transfer cost)	Bi-objective model (total transportation costs and CO_2e emissions-related costs)	• ε-constraint • weighted sum • Normalized weighted sum

Table 2 (continued)

Authors	Field/decision making problem	Disruption	CO_2/CO_2e emission optimization		Due date consideration	Performance measure(s)	Modelling approach	Solution method
			Bi-objective optimization	Charging method				
Resat and Turkay (2019)	• Route planning		✓		✓	• Transportation cost (including transfer cost, inventory holding cost, early arrival, late arrival) • Carbon emission rates	Bi-objective optimization (cost and CO_2 emission rate)	• One-direction search • Augmented • ε-constraint
Maiyar and Thakkar (2019b)	• Hub location-allocation • Route planning	Disruptions at hubs		✓		• Hub location cost • Transportation cost (including environmental and social costs)	Stochastic programming	PSODE
Maiyar and Thakkar (2019c)	• Hub location-allocation • Route planning	Disruptions at hubs		✓		• Hub location cost • Transportation cost (including environmental cost)	Robust optimization (minimization of total relative regret)	PSODE
Maiyar and Thakkar (2019a)	• Hub location-allocation • Route planning		✓		✓	• Supply network cost (including transportation cost, hub location cost, penalty of violating service time) • GHG emissions	Bi-objective model	MOPSODE NSGA II

Table 2 (continued)

Authors	Field/decision making problem	Disruption	CO_2/CO_2e emission optimization		Due date consideration	Performance measure(s)	Modelling approach	Solution method
			Bi-objective optimization	Charging method				
Kabadurmus and Erdogan (2020)	• Three- echelon Supply chain • Flow network planning	Risk of suppliers		✓	✓	• Supply chain cost (including transportation cost, purchasing/production cost, supplier /plant activation cost) • Emission cost	MILP	CPLEX
Yang et al. (2021)	• Network equilibrium					Emission value	NLP (CO_2 emission objective)	Active set algorithm (ASA)
Qi et al. (2022)	• Service selection			✓	✓	Transportation cost (including carbon emission cost, and in-transit inventory cost)	MIP	CPLEX
Fallahi et al. (2023)	• Roue planning • Transportation pricing under government intervention				✓	• Transportation cost • Network's fuel consumption rates	Stochastic NLP	GAMS
Hussein et al. (2023)	• Logistics planning • Installation		✓			• Total costs • Project duration • Carbon emissions	Simulation-based multi-objective optimization	Simulation integrated with optimization and DOE

Table 2 (continued)

Authors	Field/decision making problem	Disruption	CO_2/CO_2e emission optimization		Due date consideration	Performance measure(s)	Modelling approach	Solution method
			Bi-objective optimization	Charging method				
Current study	• Route planning • Disruption handling • Resilience • Recovery actions • Preparedness actions • Budget constraints • Responsiveness and service level	Disruptions at links by: • Bombing • Terrorist attack • Flooding	✓		✓	• Transportation cost (including transfer cost and penalty cost for unmet demand) • Resilience level • Cost of preparedness and recovery activities • Emission value	Scenario-based two-stage stochastic programming	Augmented ε-constraint

Mostly in the literature of sustainable multimodal transportation, CO_2 emissions and CO_2e emissions are considered as sustainability measures (e.g., Lee et al., 2017; Lu et al., (2019; Maiyar & Thakkar, 2019b; Mostert et al., 2017; Qu et al., 2014; Resat & Turkay, 2019; Sun et al., 2018, 2019). The CO_2e emissions factor considers total GHG emissions as carbon dioxide equivalent. Both CO_2 and CO_2e can be converted into monetary units and the associated environmental cost can be estimated by taxing the total carbon emissions. Hence, modeling this measure, as emission charges in the cost objective function is simple and has been done by several researchers (Maiyar & Thakkar 2019a; Balaman et al., 2018; Demir et al., 2019; Hrušovský et al., 2016; Maiyar & Thakkar, 2019b, 2019c; Okyere et al., 2019). However, in this approach, optimization decisions might neglect emission concerns if emission charges are insignificant in comparison to other cost terms. Consequently, it is recommended to use multi-objective modeling to handle emissions as a separate objective (Demir et al., 2015).

For the strategic planning level, Mostert et al. (2017) formulated a bi-objective location model for Belgium's road, rail, and inland waterways intermodal network, with environmental effects, in terms of CO_2e, as an objective function. Lu et al. (2019) addressed the location-routing problem with consolidation in a sustainable multimodal transportation system. Balaman et al. (2018) also focused on minimizing CO_2e as a sustainability related objective function, in addition to other objectives including (i) maximizing total profit and (ii) minimizing total transportation distance.

Few papers consider sustainability in the tactical level of planning. For example, Qu et al. (2014) included sustainability in an intermodal multicommodity network consisting of truck, train, and ship. The proposed model determines the routes, transport modes, and the flow distribution through the selected routes and modes by respecting constraints on the link capacities, flows, and demand requirements. Demir et al. (2019) also tried to combine economic and environmental criteria at tactical and operational levels. The authors proposed a green bi-objective model with the objectives of minimizing total transportation costs and CO_2e-related costs. In their case study conducted in Homburg, the terminals were assumed to function as either start or intermediate points, with one terminal designated as the destination.

Hamrouni et al. (2021) introduced a new generation of intelligent decision support systems for business model design. Their research focused on artificial intelligence contributions to sustainable development through an explainable intelligent system, aiding in the design of business models, increasing productivity, reducing production costs and emissions, decreasing resource intensity, and improving market correspondence. Tamannaei et al. (2021) employed a game-theoretic approach to analyze the competition between road and intermodal road-rail systems, incorporating government intervention. Similarly, Fallahi et al. (2023) conducted similar research on government intervention in three freight transportation networks categorized as green, semi-green, and non-green settings. They classified road transportation as a non-green mode and considered the green and semi-green settings as environmentally sustainable networks. Their suggestion of semi-green transportation involves the utilization of both green and non-green modes, referred to as intermodal transportation, to harness the advantages of each system.

Beyond the studies presented in Table 2, additional researchers have explored the relationship between freight transport, energy consumption, CO_2 emissions, and economic growth to formulate reasonable and specific transport policies. For instance, Benali and Feki (2018) conducted such an analysis in the context of Tunisia, while Peng and Wu (2019) focused on the transport sector in China.

2.3 Comparative analysis and contributions

In this section, we conduct an analysis of the literature review presented in Sects. 2.1 and 2.2 to identify research gaps and emphasize our contributions in comparison to the existing body of literature.

A comparative analysis of Table 1 reveals significant research gaps in the field of disruption management in multimodal transportation. Within the existing research landscape, only two published papers (i.e., Maiyar & Thakkar, 2019b, 2019c) stand out for incorporating both environmental issues and random disruptions concurrently in their models. However, a limitation of these models is their oversight of preparedness and recovery strategies to address disruptions. Another notable finding from Table 1 is the scarcity of studies, that successfully differentiate among various disruption types such as floods and terrorist attacks. Furthermore, the table highlights a limited body of literature on disruption risk management specifically exploring various cost components of intermodal transport. These components include the fixed and variable costs of shipping commodities, the expenses associated with implementing proactive and reactive actions, and transfer costs for commodities in terminals.

Table 2 also uncovers important research gaps in the field of sustainability in multimodal transportation that merit more attention. For instance, most of the existing models are defined in a deterministic environment, overlooking the possibility of random disruptions. Likewise, Table 2 indicates that most of the existing models fail to account for budget constraints as well as responsiveness and service level measures, such as lost sales and service time. Additionally, a considerable number of published works focus mostly on the cost objective or profit-oriented economic measures, ignoring cases where emission charges are insignificant in comparison to other cost terms and are negligible in the system costs.

Addressing these research gaps, the proposed research makes significant contributions to the literature in several ways. Firstly, we introduce a two-stage stochastic optimization

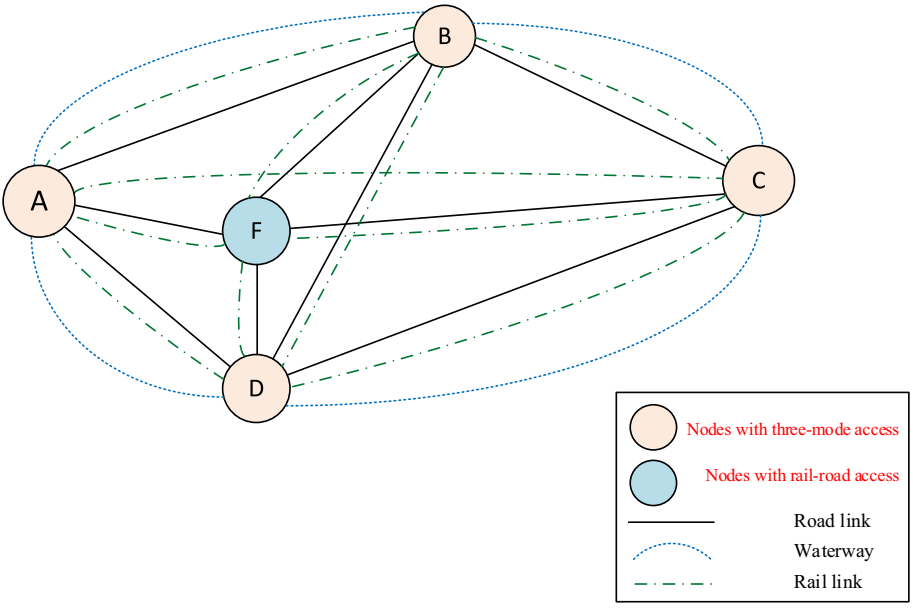


Fig. 1 Schematic of the intermodal network

approach that integrates resilience and sustainability measures by considering random disruptions and carbon emissions within the intermodal transportation network. Apart from identifying optimal proactive and reactive strategies for coping with disruptions, the proposed two-stage stochastic model determines the optimal number of vehicles from each transportation mode, the flow of containers on transportation links, and the optimal routes for order delivery. Moreover, the model's bi-objective nature enables a balance between the economic and environmental aspects of sustainability.

To offer a more comprehensive and precise economic analysis, the model incorporates various cost components of intermodal transport systems. This includes the fixed and variable costs of shipping commodities, the expenses associated with implementing proactive and reactive actions, and transfer costs for commodities in terminals. Thirdly, the model integrates different responsiveness and service level measures, such as lost sales and service time, enhancing the utility of the proposed approach. Application of this approach to a case study involving an intermodal transportation network in the UK, subject to diverse disruptions like floods and terrorist attacks, yields valuable managerial insights into the intricate interactions among resilience, sustainability, cost-efficiency, and responsiveness.

3 Problem definition and methodology

In this research, a network model of the intermodal freight transportation system with different modes of transport including truck, train, and barge is developed. The considered intermodal network is represented by directed graph G, where $N = \{1, \ldots, n\}$ is the set of nodes, and $A = \{(i,j) | (i,j) \epsilon N\}$ is the set of directed arcs (also known as links). The links represent the modal connectivity between nodes, which is different in the road, rail, and marine intermodal networks, hence, G consists of subnetworks, one for each mode. Figure 1 shows a schematic of the intermodal transportation network in which all nodes are connected by three modes of transportation except node F, which is an inland terminal with access to rail and road.

In this model, orders are known based on origin–destination (OD) city-pairs, represented by k ($k \in K$), where K is the set of all origin–destination pairs. r^k is the demand of order k, i.e., number of containers to be moved between its origin node, O(k), and destination node, D(k) (Nocera et al., 2024). These shipments can be moved using set M, which can include trucks, trains, barges, etc. represented by the set M = {truck, train, barge, etc.}. Each modal link has a definite and pre-defined capacity q_{ij}^m, which represents the number of containers that can be transported along the link $(i, j) \in A$ by mode m. Transportation cost of vehicles of mode m ($m \in M$) running on link $(i, j) \in A$ includes both fixed and variable costs, represented by f_{ij}^m and c_{ij}^m, respectively. Fixed costs are the expenses of keeping the vehicle ready for use including the cost to rent the vehicle, operator salaries, and handling costs resulted from moving containers on and off the vehicle, as elaborated by Qu et al. (2014). The variable cost per container unit of an order consists of the vehicle fuel costs, crew costs, overhead costs, and administration costs (Qu et al., 2014). A disruption may eliminate or partially decrease link capacity, thus necessitating application of actions involving preparedness (i.e., mitigation) or recovery, as defined and explained in Sect. 2.1.

In our tactical-level intermodal transportation network planning model, with CO_2e serving as the environmental indicator, we employ the methodology outlined by Demir et al. (2019) for calculating CO_2e. This method computes emission values based on the distance

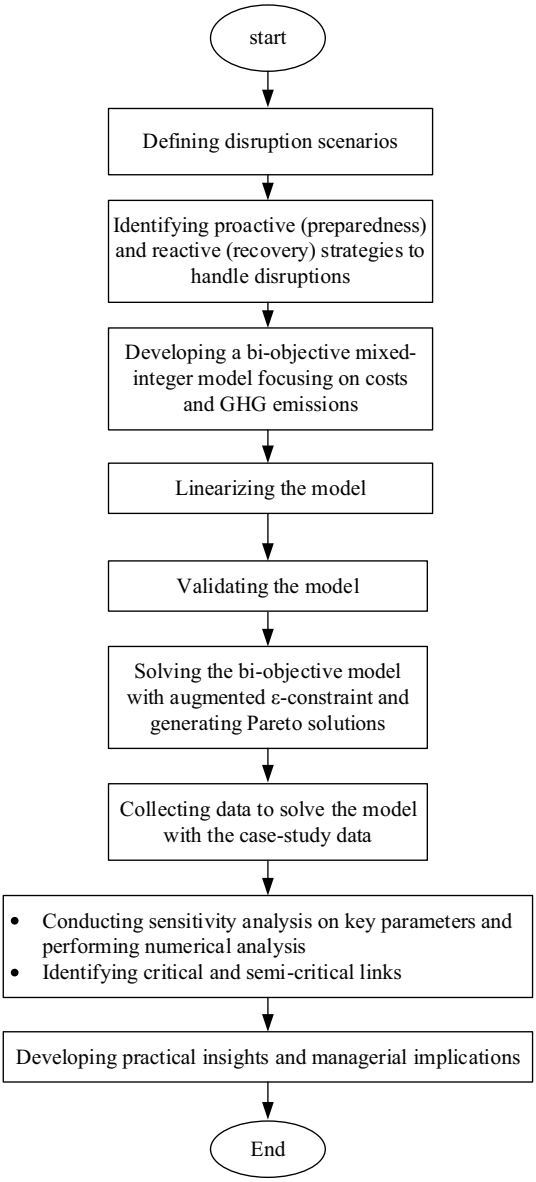


Fig. 2 The flowchart of research methodology

traveled. The primary objective of our model is to evaluate the influence of operational costs and CO_2e emissions on route selection and the allocation of flows between road and other intermodal transport modes. Furthermore, it seeks to identify the optimal set of preparedness and recovery actions to manage disruption risks.

Our model, integrating the resilience concept introduced by Chen and Miller-Hooks (2012) for intermodal transport networks, aims to strike a balance between economic and environmental objectives. The first objective is to find the (a) routes to send flows r^k between the origin and destination cities; (b) number of vehicles in each transport mode;

Table 3 Sets and parameters used in the model

Notation	Definition
N	Set of all nodes indexed by i, j, l
K	Set of all origin–destination (OD) city-pairs (order) indexed by k
M	Set of all modes including truck, train, and barge indexed by m
S	Set of disruption scenarios, indexed by s
W	Set of candidate recovery activities indexed by w
P	Set of available preparedness actions indexed by p
$O(k)$	Set of origin of orders
$D(k)$	Set of destination of orders
c_{ij}^{m}	Variable cost for shipping orders on link *(i, j)* by mode m
d_{ij}^{m}	Distance between node i and j using mode m
f_{ij}^{m}	Fixed cost of shipping orders on link *(i, j)* by mode m
b_j	Unit transfer cost in terminal j
e_j	Emissions in kg per transshipment of a container in terminal j
e_m	Emissions per transportation of a container by mode m (in kg of CO_2e/container-mile)
r^k	Quantity of order k that is to be sent from *o(k)* to *d(k)*
q_{ij}^{m}	Capacity of link *(i, j)* by mode m in normal condition (in containers)
Δq_{ij}^{mw}	Change in capacity of link *(i, j)* by mode m, if recovery activity w is taken
$\Delta' q_{ij}^{mp}$	Change in capacity of link *(i, j)* by mode m, if preparedness action p is taken
v^m	The capacity of vehicle m (in containers)
T_k	Maximum allowable shipment time for order k
τ_j	Time delay at intermodal terminal j for transferring a shipment from one mode to another mode
t_{ij}^{m}	Transit time of a shipment between i and j using mode m in normal condition
$t_{ij}'^{mw}$	Transit time of a shipment between i and j using mode m after implementing recovery activity w
o_{ij}^{mw}	Time needed to implement recovery activity w on link *(i, j)* by mode m
g_{ijs}^{m}	Percentage of capacity of link (i, j) by mode m disrupted under scenario s
δ_{ijs}^{m}	Percentage added to transit time of a shipment between i and j by mode m due to disruption in scenario s
u_k	Unit cost of lost sales for order k (£/container)
a_{ij}^{mw}	Cost of implementing recovery activity w on link (i, j) for mode m
$\tilde{a}_{ij}^{mpw}$	Cost of implementing recovery activity w on link (i, j) for mode m, if preparedness action p is taken
h_{ij}^{mp}	Cost of implementing preparedness action p on link (i, j) for mode m
B_s	Maximum budget for recovery and preparedness activities in scenario s
π_s	Possibility of occurrence of scenario s
λ	Matrix of preparedness-recovery activity relationship in which each element λ_{pw} is set to 1 if recovery activity w is affected by preparedness action p, and 0 otherwise

and (c) set of preparedness and recovery actions to implement, such that the expected total of freight transportation cost, cost of implementing recovery and preparedness activities, and lost sales cost is minimized. The second objective focuses on CO_2e, both in the transportation from origin to destination and in the transshipment process in the terminals. The

Table 4 Decision variables used in the model

Notation	Definition
X_{ijs}^{km}	Flow of order k on link (i, j) using mode m in scenario s (in container)
n_{js}^{k}	Number of containers of order k transshipped at terminal j in scenario s
Y_{ij}^{m}	Number of vehicles of mode m transported on link (i, j)
Z_{ijs}^{km}	Binary variable, equal to 1 if a shipment for order k passes through link (i, j) using mode m in scenario s; 0, otherwise
U_{s}^{k}	Number of shipments that cannot be satisfied for order k under scenario s
γ_{ijs}^{mw}	Binary variable, equal to 1 if recovery activity w is undertaken on link (i, j) by mode m in scenario s; 0, otherwise
β_{ij}^{mp}	Binary variable, equal to 1 if preparedness activity p is undertaken on link (i, j) by mode m; 0, otherwise

model accounts for link capacity, time requirement of orders (under both normal and disruption conditions), and budget constraints.

Building upon the sustainable model proposed by Qu et al. (2014), we formulate the problem as a two-stage stochastic mixed-integer programming (SMIP) model comprising two types of variables: those in the strategic level of decision making and determined in the first stage, while the others, called second stage variables, are decided based on the disruption scenario. In the real world, the vehicle fleet to move shipments must be pre-planned, so number of vehicles from each transport mode over link (i, j), y_{ij}^{m}, is decided at the first stage.

Figure 2 depicts the procedural steps of the proposed methodology. After outlining the research assumptions, the following sections present more details for each step.

3.1 Assumptions

Utilizing the notations introduced in Tables 3 and 4, we establish the following assumptions to precisely delineate the problem's boundaries and scope.

- We assume that shipments travel in freight containers and can be transferred between different modes. Intermodal transfer cost is incurred when a shipment changes mode from one transportation mode to another within an intermodal terminal.
- At an intermodal terminal, a delay arises from transferring freight between two different modes.
- When mode m for link (i, j) is disrupted under scenario s, its associated capacity, q_{ij}^{m}, is reduced to $q_{ij}^{m}\left(1 - g_{ijs}^{m}\right)$, where g_{ijs}^{m} represents the percentage of the link's capacity that is disrupted. If link (i, j) by mode m is fully disrupted, its associated capacity, q_{ij}^{m}, becomes 0, meaning that mode m is no longer available on link (i, j) (e.g., the rail link between nodes i and j has collapsed).
- In disruption scenario s, the shipment transit time over a link (i, j) increases from t_{ij}^{m}, in normal conditions, to $t_{ij}^{m}\left(1 + \delta_{ijs}^{m}\right)$, where δ_{ijs}^{m} is the percentage added to the normal

transit time due to a disruption. The implementation of a recovery action can mitigate the resulting transit time on a link, reducing it to t'^{mw}_{ij}.

- Both recovery and preparedness actions can be implemented on each link to deal with disruptions.
- We have incorporated a maximum allowable shipment time for each order, aligning it with the due date/delivery time specified by customers. This due date consideration is a common practice in customer orders and is crucial in real-world freight transport networks. Failing to meet these due dates can significantly impact customer satisfaction and lead to lost demand (Hussein et al., 2023; Sun et al., 2021; Wang et al., 2021; Zhao et al., 2018).

3.2 Mathematical model

Expanding upon the sustainable model introduced by Qu et al. (2014), we have crafted a two-stage stochastic programming model aimed at deriving optimal planning decisions. Our two-stage mixed-integer program formulation is presented in (1) – (15). The first-stage variables (e.g., preparedness activities on critical links) are preserved in all disruption scenarios. The second-stage variables (flow of shipments along each link for each order, number of containers transshipped at intermodal terminals, unsatisfied demand, and the set of recovery activities) are tuned to the specific disruption scenarios.

The formulation contains integer variables (e.g., number containers and mode-specific vehicles); continuous variables (e.g., shipment flows along each link for each order, and unsatisfied demand for each order); and binary variables (e.g., the selection of recovery and preparedness activities on corresponding links, and the selection of links carrying flow).

$$min \sum_m \sum_i \sum_j f^m_{ij} Y^m_{ij} + \sum_i \sum_j \sum_m \sum_p h^{mp}_{ij} \beta^{mp}_{ij} + \sum_s \pi_s \left(\sum_k \sum_m \sum_i \sum_j c^m_{ij} X^{km}_{ijs} + \sum_k \sum_j b_j n^k_{js} + \sum_i \sum_j \sum_m \sum_w a^{mw}_{ij} \gamma^{mw}_{ijs} + \sum_k u_k U^k_s \right) \tag{1}$$

$$min \sum_s \pi_s \left(\sum_k \sum_m \sum_i \sum_j X^{km}_{ijs} d^m_{ij} e_m + \sum_k \sum_j e_j n^k_{js} \right) \tag{2}$$

Subject to

$$\sum_j \sum_m X^{km}_{ijs} - \sum_l \sum_m X^{km}_{lis} = \begin{cases} r^k - U^k_s; & if\ i = O(k),\ \forall k \in K,\ \forall s \in S \\ -r^k + U^k_s; & if\ i = D(k),\ \forall k \in K,\ \forall s \in S \\ 0; & otherwise \end{cases} \tag{3}$$

$$\sum_k X_{ijs}^{km} \le \left(1 - g_{ijs}^m\right) q_{ij}^m + \sum_w \gamma_{ijs}^{mw} \Delta q_{ij}^{mw} + \sum_p \beta_{ij}^{mp} \Delta' q_{ij}^{mp}; \quad \forall (i,j) \in A,\ \forall m \in M,\ \forall s \in S \tag{4}$$

$$\sum_k X_{ijs}^{km} \le Y_{ij}^m v^m; \forall (i,j) \epsilon A, \forall m \epsilon M, \forall s \epsilon S \tag{5}$$

$$\sum_i \sum_j \sum_m Z_{ijs}^{km} t_{ij}^m \left(1 + \delta_{ijs}^m\right) + \sum_i \sum_j \sum_m \sum_w Z_{ijs}^{km} \gamma_{ijs}^{mw} \left(t'^{mw}_{ij} - t_{ij}^m \left(1 + \delta_{ijs}^m\right)\right) + \sum_i \sum_j \sum_m \sum_w \gamma_{ijs}^{mw} o_{ij}^{mw} + \sum_j \tau_j n_{js}^k \le T_k; \forall k \in K, \forall s \in S \tag{6}$$

$$\sum_i X_{ijs}^{km} - \sum_l X_{jls}^{km} \le n_{js}^k; \quad \forall k \in K,\ j \in N \backslash D(k),\ \forall m \in M,\ \forall s \in S \tag{7}$$

$$Z_{ijs}^{km} \le X_{ijs}^{km}; \forall (i,j) \epsilon A, \forall k \epsilon K, \forall m \epsilon M, \forall s \epsilon S \tag{8}$$

$$X_{ijs}^{km} \le r^k Z_{ijs}^{km}; \forall (i,j) \epsilon A, \forall k \epsilon K, \forall m \epsilon M, \forall s \epsilon S \tag{9}$$

$$\sum_i \sum_j \sum_m \sum_p h_{ij}^{mp} \beta_{ij}^{mp} + \sum_i \sum_j \sum_m \sum_w a_{ij}^{mw} \gamma_{ijs}^{mw} + \sum_i \sum_j \sum_m \sum_p \sum_w (\tilde{a}_{ij}^{mpw} - a_{ij}^{mw}) \lambda_{pw} \beta_{ij}^{mp} \gamma_{ijs}^{mw} \le B; \forall s \epsilon S \tag{10}$$

$$\sum_w \gamma_{ijs}^{mw} \le 1; \forall (i,j) \epsilon A, \forall m \epsilon M, \forall s \epsilon S \tag{11}$$

$$\sum_p \beta_{ij}^{mp} \le 1; \forall (i,j) \epsilon A, \forall m \epsilon M \tag{12}$$

$$n_{js}^k, Y_{ijs}^m \epsilon \{0,1,2..\} \forall (i,j) \epsilon A, \forall m \epsilon M, \forall k \epsilon K, \forall s \epsilon S \tag{13}$$

$$X_{ijs}^{km} \ge 0; U_s^k \ge 0 \forall (i,j) \epsilon A, \forall m \epsilon M, \forall k \epsilon K, \forall s \epsilon S \tag{14}$$

$$Z_{ijs}^{km}, \gamma_{ijs}^{mw}, \beta_{ij}^{mp} \epsilon \{0,1\} \forall (i,j) \epsilon A, \forall m \epsilon M, \forall k \epsilon K, \forall w \epsilon W, \forall p \epsilon P, \forall s \epsilon S \tag{15}$$

The expressions in (1) and (2) respectively, show the functions for the two objectives: cost minimization and emission minimization. Flow conservation restrictions are in Constraint (3), which also expresses the demand of orders and the quantity of lost sales. As can be inferred, each order has only one origin and one destination. Constraint (4) formulates the capacity constraint of each link after disruption occurrence and applying a preparedness and a recovery activity. This constraint guarantees that the total flow on link *(i, j)* using mode *m* to be less than the capacity of the modal link under different scenarios. Constraint (5) determines the best number of required vehicles on mode *m*. Constraint (6) requires that the total shipment time of route used for order *k* not exceed

a given maximum duration T_k. This constraint aligns with the due date time specified by customers. Failing to meet these due dates can significantly impact customer satisfaction and lead to lost demand (Hussein et al., 2023; Sun et al., 2021; Wang et al., 2021; Zhao et al., 2018). Constraint (6) accounts for transfer time for mode changes and shipment time changes resulting from both the impacts of disruptive events and the implementation of recovery actions. Constraint (7) identifies if a transfer operation at a particular terminal j is conducted, and it calculates the number of transferred containers. Since splitting shipments is allowed, it is possible to change partially the mode of containers in a shipment.

Constraints (8) and (9) define the binary variable Z_{ij}^{km} with respect to the value of X_{ij}^{km}. Constraint (10) ensures that the total cost of the selected preparedness and recovery actions does not exceed a given budget under each scenario, while Constraint (11) requires that only one recovery activity, from a set of recovery actions, can be chosen for each link. Constraint (12) is similar to Constraint (11), but it refers to preparedness activities. Both Constraints (11) and (12) ensure that conflicting actions will not be simultaneously selected. The domains of the decision variables are defined in constraints (13) – (15).

3.3 Linearization

Constraints (6) and (10) are nonlinear because of the nonlinear terms of $Z_{ijs}^{km}\gamma_{ijs}^{mw}$ in Constraint (6) and $\beta_{ij}^{mp}\gamma_{ijs}^{mw}$ in Constraint (10), in which two binary variables are multiplied. Based on the linearization technique described in Williams (2013), we define $Z_{ijs}^{km}\gamma_{ijs}^{mw}$ as $\mathcal{R}_{ijs}^{kmw} \in \{0,1\}$ and $\beta_{ij}^{mp}\gamma_{ijs}^{mw}$ as $\psi_{ijs}^{mpw}\epsilon\{0,1\}$. The following Eqs. (16) – (19) demonstrate the linearization process of term $Z_{ijs}^{km}\gamma_{ijs}^{mw}$ in Constraint (6).

$$Z_{ijs}^{km} + \gamma_{ijs}^{mw} - \mathcal{R}_{ijs}^{kmw} \leq 1; \forall (i,j) \in A,\ \forall m \in M,\ \forall k \in K,\ \forall w \in W,\ \forall s \in S \tag{16}$$

$$\mathcal{R}_{ijs}^{kmw} \leq \gamma_{ijs}^{mw}; \forall (i,j) \in A,\ \forall m \in M,\ \forall k \in K,\ \forall w \in W,\ \forall s \in S \tag{17}$$

$$\mathcal{R}_{ijs}^{kmw} \leq Z_{ijs}^{km}; \forall (i,j) \in A,\ \forall m \in M,\ \forall k \in K,\ \forall w \in W,\ \forall s \in S \tag{18}$$

$$\mathcal{R}_{ijs}^{kmw} \geq 0; \forall (i,j) \in A,\ \forall m \in M,\ \forall k \in K,\ \forall w \in W,\ \forall s \in S \tag{19}$$

The following Eqs. (20)–(23) also show the linearization of $\beta_{ij}^{mp}\gamma_{ijs}^{mw}$ in Constraint (10).

$$\beta_{ij}^{mp} + \gamma_{ijs}^{mw} - \psi_{ijs}^{mpw} \leq 1; \forall (i,j)\epsilon A, \forall m\epsilon M, \forall p\epsilon P, \forall w\epsilon W, \forall s\epsilon S \tag{20}$$

$$\psi_{ijs}^{mpw} \leq \gamma_{ijs}^{mw}; \forall (i,j)\epsilon A, \forall m\epsilon M, \forall p\epsilon P, \forall w\epsilon W, \forall s\epsilon S \tag{21}$$

$$\psi_{ijs}^{mpw} \leq \beta_{ij}^{mp}; \forall (i,j)\epsilon A, \forall m\epsilon M, \forall p\epsilon P, \forall w\epsilon W, \forall s\epsilon S \tag{22}$$

$$\psi_{ijs}^{mpw} \geq 0; \forall (i,j) \epsilon A, \forall m \epsilon M, \forall p \epsilon P, \forall w \epsilon W, \forall s \epsilon S \tag{23}$$

As a result, instead of Constraints (6) and (10), the following constraints would be substituted into the model.

Constraints (16) – (23)

$$\sum_i \sum_j \sum_m Z_{ijs}^{km} t_{ij}^m \left(1 + \delta_{ijs}^m\right) + \sum_i \sum_j \sum_m \sum_w \mathcal{R}_{ijs}^{kmw} \left(t_{ij}^{\prime mw} - t_{ij}^m \left(1 + \delta_{ijs}^m\right)\right) + \sum_i \sum_j \sum_m \sum_w \gamma_{ijs}^{mw} o_{ij}^{mw} + \sum_j \tau_j n_{js}^k \leq T_k; \forall k \in K, \forall s \in S \tag{24}$$

$$\sum_i \sum_j \sum_m \sum_p h_{ij}^{mp} \beta_{ij}^{mp} + \sum_i \sum_j \sum_m \sum_w a_{ij}^{mw} \gamma_{ijs}^{mw} + \sum_i \sum_j \sum_m \sum_p \sum_w \left(\tilde{a}_{ij}^{mpw} - a_{ij}^{mw}\right) \lambda_{pw} \psi_{ijs}^{mpw} \leq B; \forall s \in S \tag{25}$$

3.4 Solution method: augmented ε-constraint technique

Multi-objective optimization seeks to optimize multiple conflicting objectives simultaneously. In this type of problem, a single solution cannot guarantee simultaneous optimization of all objectives, meaning that a typical solution may improve an objective function, while worsening others. This necessitates trade-offs in satisfying different objectives. A solution is called non-dominated, if cannot be dominated by another solution which improves one objective function without worsening another (Demir et al., 2019).

Solution approaches for multi-objective problems include goal programming, weighted sums, and ε-constraint: an efficient technique (Bérubé et al., 2009) to provide an approximation of the Pareto set, which is sufficient in most cases (Mavrotas & Florios, 2013). This method is widely used in multi-objective studies due to its simplicity and applicability for convex and nonconvex problems, allowing for the generation of a set of non-dominated solutions and trade-offs among the objective functions (Allehashemi et al., 2022). Unlike most other techniques, ε-constraint obviates the need to scale the objective functions by assigning weights to the objectives (Jabbarzadeh et al., 2020), which significantly affect the solution in techniques such as goal programming and weighted sum approaches. A more advanced variant of ε-constraint is the AUGCON (augmented ε-constraint), introduced by Mavrotas (2009), which offers the advantage of controlling the number of non-dominated solutions. This approach, known for its simplicity and applicability, has been reported as a successful strategy in several papers, such as Hartillo-Hermoso et al. (2020). The AUGCON method is able to generate non-extreme solutions, while methods such as weighted sum and normalized weighted sum can produce one extreme solution in each run (Demir et al., 2019).

Consider a bi-objective problem in the form of:

$$Min\{f_1(x), f_2(x)\} \tag{26}$$

Subject to

$$x \epsilon X \tag{27}$$

where X is the feasible region of the program. In the augmented ε-constraint technique, one of the objective functions is optimized, while all others are converted into constraints. To implement this technique in a bi-objective model with minimization objectives, the following general steps are involved (Allehashemi et al., 2022).

1- Firstly, each objective function is individually minimized to obtain the best (referred to as utopia) and the worst (referred to as nadir) points. For instance, $f_1(x)$ is minimized to find the ${f_1}^{Best}$ and ${f_2}^{Worst}$ values, and then, $f_2(x)$ is minimized to find the ${f_1}^{Worst}$ and ${f_2}^{Best}$ values. In minimization problems, the optimal values represent the lower bounds (utopia points) of each objective.
2- Subsequently, a value for } (the number of grid points) is set, representing the number of intervals for the objective function that is moved to the constraints set, resulting in } + 1 possible non-dominated solutions. The number of iterations is, therefore, }+1. Additionally, a value for ε *(eps)* is set, and a new non-negative continuous variable s_2 is defined for the objective function that will be included in the constraints.
3- The Model B (AUGCON) is then ready to be executed with a single objective function as the main objective, as outlined below:

Model B (AUGCON):

$$Minf_1(x) - eps \times \frac{s_2}{f_2^{Worst} - f_2^{Best}} \tag{28}$$

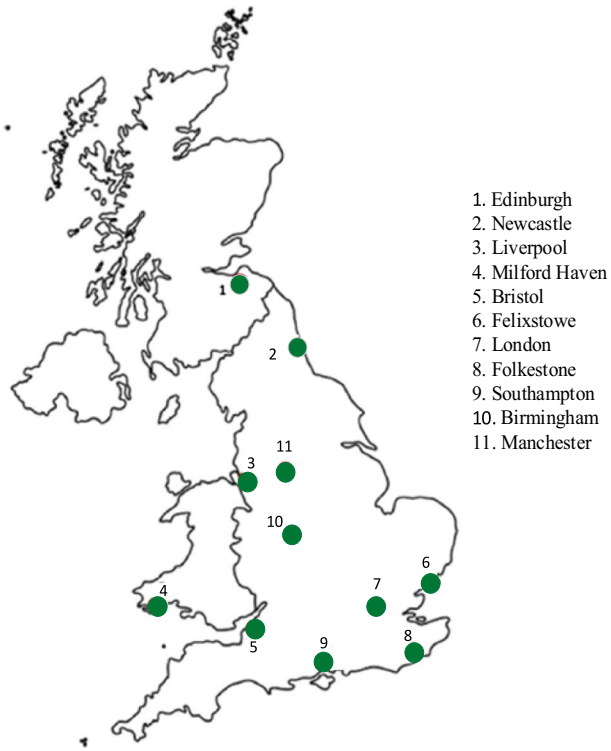


Fig. 3 The UK cities and ports in the network (adapted from Qu et al. (2014))

Subject to:

$$x \epsilon X \tag{29}$$

$$f_2(x) = f_2^{Worst} - (iter - 1) \times \frac{f_2^{Worst} - f_2^{Best}}{\mathscr{J}} - s_2 \tag{30}$$

$$s_2 \geq 0 \tag{31}$$

where *iter* represents the iteration index, and f_2^{Worst} and f_2^{Best} denote the nadir and utopian for the second objective function, respectively.

Table 5 Case study data on origins, destinations and demands

Order No	Origin node	Destination node	Required demand (ton)	Required demand (TEU container)	T_k (hour)
1	4	3	212	15	70
2	9	7	1182	84	80
3	2	9	794	57	150
4	4	11	128	9	150
5	2	7	182	13	150
6	11	5	99	7	110
7	10	3	150	11	70
8	10	1	343	25	200
9	11	7	168	12	70
10	3	6	567	41	230
11	7	8	960	69	40
12	1	3	240	17	230
13	2	11	790	56	110
14	5	10	83	6	110
15	5	6	570	41	190
16	5	8	150	11	160
17	11	2	1220	87	110
18	8	6	110	8	40
19	6	5	350	25	190
20	7	5	410	29	200
21	3	2	300	21	260
22	4	2	130	9	260
23	9	5	225	16	120
24	4	10	850	61	130
25	5	10	500	36	110
26	2	1	275	20	40
27	6	11	150	11	110
28	8	9	800	57	40
29	4	3	175	13	70
30	8	7	87	6	40

Table 6 Case study parameters on capacity, cost, and emissions of transport modes

Transportation mode	Capacity (TEU container)	variable cost (£ per TEU-mile)	Transfer cost (£ per TEU)	CO_2e (kg/TEU-mile)
Truck	2	0.504	19.474	1.40
Rail	28	0.595	19.474	0.50
Ship	212	0.35	19.474	0.36

4 Computational experiments

In this section, the proposed approach is implemented using the data from a case study of the UK transportation network. After conducting the numerical experiments, the results are analyzed to draw important managerial insights. The computational testing for the proposed model is performed using the Gurobi Optimizer version 9.1.0 coded in Gurobi Python API (Python 3.7) on an Asus Zenbook laptop computer with Intel Core i7 CPU and 16G RAM. The computational time for running the problems with different ε values ranges from just over a few seconds to 30 min. Prior to conducting the experiments, we validated the model by comparing its computational output with that of Qu et al. (2014) using the same data set. To run AUGCON method, *eps* value is set to 10^{-4}, as recommended by Mavrotas (2009), who suggests setting the *eps* value between 10^{-3} and 10^{-6}. Additionally, the number of grid points for the second objective function ($\}$) is also set to 19. We should note that changing the *eps* value or other parameters may yield different outputs and result in a different Pareto set.

4.1 Description of the case study

The application of the proposed approach is investigated using the data of a case study in the UK. The network has 11 nodes which include nine ports (Edinburgh, Newcastle, Liverpool, Milford Haven, Bristol, Felixstowe, London, Folkestone, and Southampton) and two inland cities (Birmingham and Manchester), which are connected to other nodes by highways and railways. For the other nodes, three transportation modes (truck, train, and barge) are available. Therefore, the network includes 292 links (11×10 for highways; 11×10 for railways; and 9×8 for seaways). Figure 3 shows the geographical distribution of the nodes in the network.

We selected the UK transport network as our validation dataset for several reasons. Firstly, although the network comprises only 11 nodes, it includes a substantial number of

Table 7 The summary of scenarios and their probabilities

Scenario category	Total probability	Number of scenarios
Baseline	0.772	1
Flood	0.18	7
Terrorist attacks	0.024	12
Bombing	0.024	6
Total	1	26

links totaling 292, which covers a significant portion of the entire freight transport network in the UK. This extensive network provides a diverse and comprehensive dataset for our analysis, encompassing highways, railways, and seaways for freight transport. Moreover, the interconnected nature of the majority of links via multiple modes of transport in the UK facilitates seamless transfer operations, making it highly suitable for our analysis. Furthermore, the UK offers readily available and acceptable quality transportation data compared to other regions, contributing to a more robust validation process. The attention this network has received from prior researchers, such as Qu et al. (2014), and Monios (2015) and Monios (2016) provides us with the opportunity to compare our findings with existing literature. The insights and recommendations derived from our study and subsequent comparison are not only relevant to the UK but also applicable to similar transportation systems worldwide.

In line with Qu et al. (2014), our network consists of 30 different orders requiring transportation. The origin–destination pairs and demand requirements for each order are specified in Table 5. The last column of Table 5 indicates the maximum allowable time for fulfilling each order.

Table 6 shows the data for capacity, variable cost, transfer cost, and CO_2e emissions factor of each mode, based on Qu et al. (2014).

An average weight of 14 ton/TEU is assumed (Demir et al., 2019). To estimate CO_2e emissions, we assume that each TEU produces as much as 11.27 kg of CO_2e emissions at a terminal (Budiyanto et al., 2019).

4.2 Designing disruption scenarios

To reflect disruption scenarios relevant to the UK, we collected and analyzed data associated with the various disruptions (floods, bombings, and terrorist attacks). The UK transportation network is subject to floods, terrorist attacks, and bombing, but rarely encounters freight flow interruptions from earthquakes. Using the historical data, we identified 26 probable scenarios in three disruption categories (flood, terrorist attacks, and bombing) and in the baseline scenario. The occurrence probability of a disruption scenario for the transportation network can be estimated by the fraction of time during a year in which a transportation link (or a set of links) are disrupted (Jabbarzadeh et al., 2020). For instance, if the historical data shows that a link was non-functional for 10 of the 365 days due to disruption, then the probability of disruption for that link would be 2.7% (i.e., 10/365). The summary of scenario categories and their probabilities are presented in Table 7.

Table 8 The frequency of each human-made disaster, in the last three decades

Decade	Man-made disaster	Total frequency
2000s	Terrorist attack	3
	Bombing	4
2010s	Terrorist attack	11
	Bombing	3
2020s	Terrorist attack	2
	Bombing	-
Total	23	

4.2.1 Flood

The flood-related disruptions are more probable than the other disruptions in the UK and this area is likely to be one of the worst flood-affected locations, according to the IPCC report. For example, between November 2019 and February 2020, severe winter flooding occurred across the UK.

Based on a report by Lake Street Consulting (Jackson & Finney, 2016), on average, about 11 floods and storms happen in a year, each of which has an impact of about 6 days, based on historical data in the National Risk Register Of Civil Emergencies (Cabinet Office, 2017), meaning that during or after each flood, it takes 6 days to return to normal condition. $(11 \times 6)/365 = 18\%$, thus, totally 18% of all days in a year are disrupted because of floods in the UK. This 18% is spread between different locations of the country.

Regarding some recent storms and floods in the UK, such as Storm Ciara in Scotland, and Storm Dennis in the entire UK, both in 2020, we gathered data from resources such as Loveproperty and Coffey (2022), which shows the ten most flood-prone locations in the UK. We assume that the probability of a flood in these regions is 1.5 of other regions such as Southampton. A flood may affect several nodes of the network, at the same time. Hence, 7 scenarios are designed to consider totally 18% of disruption probability in the UK freight network. In some cases, it may take weeks or even months to repair flood-damaged rail links, though we consider an average period of 6 days.

4.2.2 Terrorist attack and bombing

In the last decade (the 2010s), totally 14 terrorist attacks and bombings occurred, which means that on average 1.4 terrorist incidents in a year. It is very difficult to estimate the required time to return to normal condition, since different incidents have different consequences, and there is no available data in this regard. Inspiring by Chen and Miller-Hooks (2012), the difference between bombing and terrorist attack is mentioned here:

- In bombing, random links from rail and roadways are selected. If it is a railway, it will be non-functional, while highways will have reduced capacity (e.g., 30%). Seaways are totally functional.
- In the terrorist attack, a terrorist incident in random locations will decrease capacity more significantly at links near the emergency scene (e.g., 50%) than at links farther away (e.g., 20%). There will be no impact on seaways, even when a port is disrupted since it will be regarded in a separate case in future research.

Based on this definition, the frequency and type of each man-made disaster in the UK, in the last three decades are listed in Table 8.

Some examples of terrorist attacks in the UK can be named as terrorist attacks in London and Glasgow, and several bridge attacks in London. Some bombing incidents in the past 20 years include bombing at Manchester, parcel bombing at London and Glasgow, and parcel bombing at the same time to army at Oxford, Brighton, Canterbury, Slough to Reading, Aldershot, and Chatham.

In bombing, the infrastructure damage is probable, so returning to normal condition takes more time in comparison with other terrorist attacks such as stabbings or car attacks. In the case of a terrorist attack, returning to the normal condition could take about 9 days, on average, hence, $9 \times 0.98/365 = 2.4\%$. However, in the case of

Table 9 Disruption scenarios

No	Description	Probability	Type of scenario
1	Normal	0.772	Baseline
2	Flood in Southampton; all rail links to/from Southampton are non-functional	0.02	Flood
3	Flood is Edinberg and Newcastle; all rail links between these two cities are non-functional	0.03	
4	Flood in Liverpool, Birmingham and Manchester; all rail links between these three cities are non-functional	0.03	
5	Flood in Bristol; all rail links to/from Bristol are non-functional	0.03	
6	Flood in Milford; all rail links to/from Milford are non-functional	0.03	
7	Flood in Felixstowe, London, Folkestone; all rail links between these three cities are non-functional	0.02	
8	Flood in Southampton and Bristol; all rail links between these two cities are non-functional	0.02	
9	London is attacked by terrorists, just road capacity to/from London is reduced by 0.2	0.002	Terrorist attack
10	London is attacked by terrorists, just road capacity to/from London is reduced by 0.3	0.002	
11	London is attacked by terrorists, just road capacity to/from London is reduced by 0.4	0.002	
12	London is attacked by terrorists both rail and road are disrupted. Rail links to London are non-functional. Road link is reduced in capacity by 0.2	0.002	
13	London is attacked by terrorists both rail and road are disrupted. Rail links to London are non-functional. Road link is reduced in capacity by 0.3	0.002	
14	London is attacked by terrorists both rail and road are disrupted. Rail links to London are non-functional. Road link is reduced in capacity by 0.4	0.002	
15	Birmingham is attacked by terrorists and road are disrupted. Road links to/from Birmingham is reduced in capacity by 0.5	0.002	
16	Manchester is attacked by terrorists and road links are disrupted. Road links to/from Manchester is reduced in capacity by 0.5	0.002	
17	Edinburgh is attacked by terrorists and road are disrupted. Road links to/from Manchester is reduced in capacity by 0.5	0.002	
18	Oxford is attacked by bomb, so road links between London and Birmingham decrease in capacity by 0.5	0.002	
19	Folkestone is attacked by terrorists and road are disrupted. Road links to/from Folkestone is reduced in capacity by 0.5	0.002	
20	Southampton is attacked by terrorists and road are disrupted. Road links to/from Southampton is reduced in capacity by 0.5	0.002	

Table 9 (continued)

No	Description	Probability	Type of scenario
21	Link is attacked by bomb, just rail links between London and Bristol are non-functional	0.004	Bombing
22	Link is attacked by bomb, just rail links between London and Felixstowe are non-functional	0.004	
23	Link is attacked by bomb, just rail links between London and Folkestone are non-functional	0.004	
24	Link is attacked by bomb, just rail links between London and Southampton are non-functional	0.004	
25	Link is attacked by bomb, just rail links between London and Birmingham are non-functional	0.004	
26	Link is attacked by bomb, just road links between London and Birmingham decrease in capacity by 0.3	0.004	

bombing, returning to normal condition could take about 21 days to repair the infrastructure, thus, $21 \times 0.42/365 = 2.4\%$. So, these two types of scenarios have a total occurrence probability of 4.8%. The frequency of events shows that booming in rail and road links is less probable, but in comparison to other terrorist attacks, have more effect on transportation networks. Considering the above computation, the baseline probability is 77.2%.

To identify the locations more prone to terrorist incidents, we considered the frequency of different incidents in UK cities over the last three decades. The data show that 70% of these events happened in London. For the remaining 30%, the disruption scenarios are designed to include the probability of terrorist attacks in other locations such as Birmingham, Manchester, Edinburgh, Oxford, Folkestone, and Southampton. Based on the above-mentioned justifications, details of disruption scenarios are presented in Table 9.

4.3 Interaction between cost-efficiency and greenness

We first address the trade-off between total cost and total emissions. For this purpose, we solve the problem using the augmented ε-constraint method. To obtain the emission-cost trade-off, a Pareto frontier for the problem is depicted in Fig. 4.

As can be inferred from Fig. 4, moving towards greenness increases the cost. At the extreme condition, when the system is fully green, the total cost is £2,619,000. The minimum cost of this Pareto front is £86,971, with a total emission of 74,354 kg.

In order to analyze more, three non-dominated solutions are chosen, as implemented by Jabbarzadeh et al. (2020). The maximum value of ε for the second objective can be obtained by solving a single-objective model with only a total cost objective, i.e., equivalent to *Min Cost* solution. Conversely, the *Min Emission* solution is obtained by ignoring the cost objective, leading to a zero-emission solution in which all demands are lost sales. One *intermediate* solution is also chosen in which there is a trade-off between total cost and emission. *Min Emission* solution represents a system in a fully "green" state, while *Min Cost* solution is associated with the term of "brown", versus "green". The three points (middle and extreme points of Pareto front) are shown in red color Fig. 4.

Using the regression analysis, two slopes can be identified for the trendline of Fig. 4, one from *Min Cost* solution to *Intermediate* solution, and the other from *Intermediate to Min Emission* solution. The analysis shows that the slope of the first part of the trend is −0.043, while the slope of the second part is −0.018. This means that increasing transportation cost in an economically efficient and brown system (*Min Cost* solution) toward a partially green system (*Intermediate* solution) can yield a considerable decrease (about 62%) in emissions. Though, moving from a partially green system (*Intermediate solution*) towards a fully green system imposes more cost on the system. In other words, for a brown but efficient system, a slight initial increase in transportation costs can cause a considerable jump towards sustainability.

4.4 Interaction between resilience and greenness

To study the resilience-sustainability relationship, the percentage of satisfied demands has been chosen as the measure. Figure 5 shows the relationship between greenness, the total cost of preparedness and recovery activities as well as the percentage of satisfied demand.

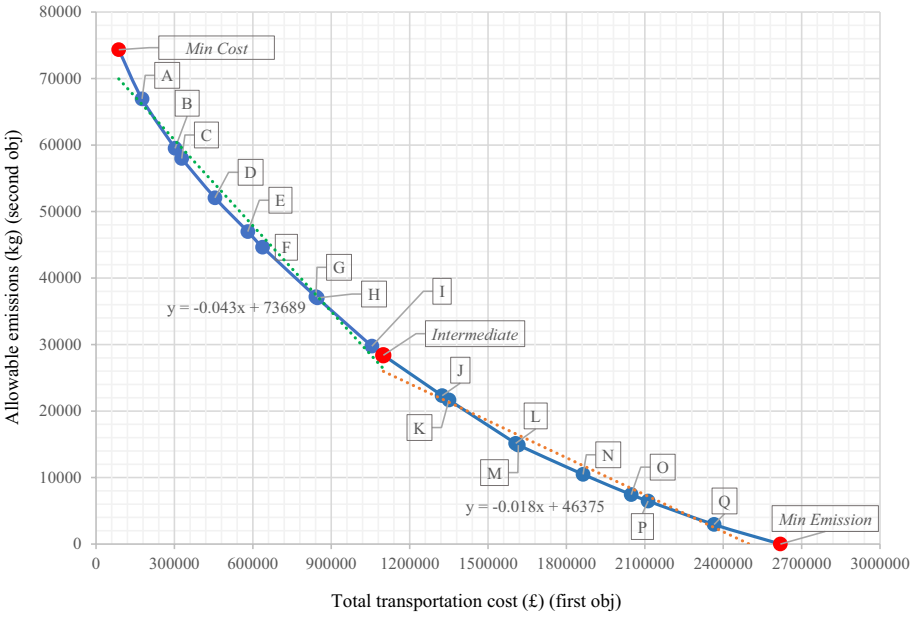


Fig. 4 Pareto front based on the interaction between cost-efficiency and greenness

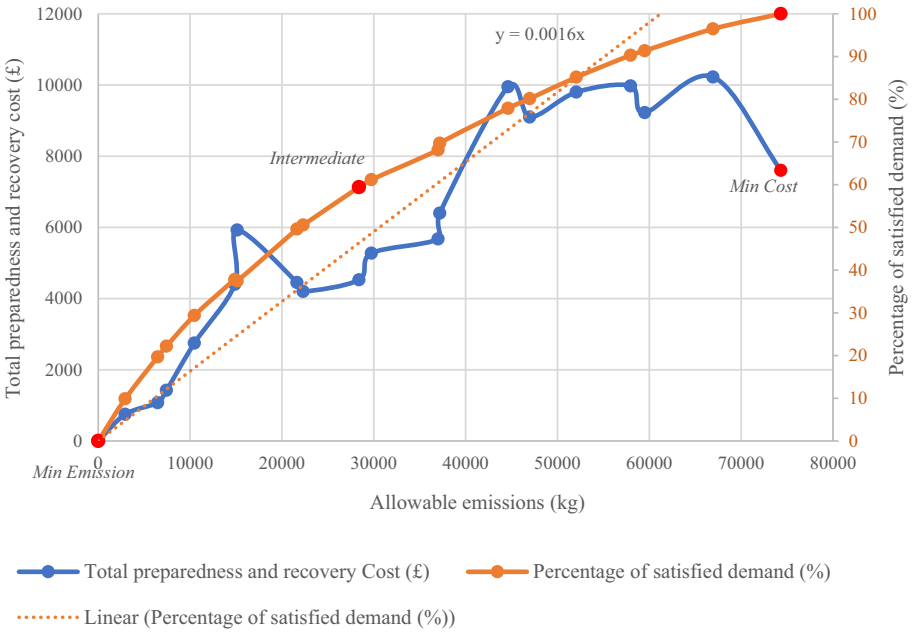


Fig. 5 Emission (kg) versus total preparedness and recovery cost (£) and satisfied demand (%)

As can be seen in Fig. 5, by increasing the threshold of CO_2e emissions (ε values), the percentage of satisfied demand increases to the extreme point of 100% for the *Min Cost* solution. In the *Intermediate* solution, about 59.4% of demands could be satisfied. To meet more than 90% of demand, at least about 57,974 kg CO_2e would be emitted, which is 78% of the maximum CO_2e emission in the system. On the other hand, by increasing the allowable CO_2e emissions, the cost of preparedness and recovery generally grows as well, though its trend fluctuates in some points.

To better see the trend, in Fig. 6, the total transportation cost and the total cost of preparedness and recovery activities are depicted for different values of allowable emissions. By increasing the allowable emissions, these two types of cost seem to have an inverse relationship. Higher values of the total cost of preparedness and recovery actions can lead to a decrease in the total transportation cost of the system, meaning that by investing in preparedness and recovery actions more disruptive situations could be handled.

4.5 Critical and semi-critical links in the network

Based on the results, the critical and semi-critical links are identified to implement preparedness activities. Critical links are those which are chosen by the model in all Pareto solutions with different values of allowable emissions (i.e., different values of ε). While, semi-critical links are identified in one or two solutions of the Pareto front, meaning that semi-critical links appear in some degree of system greenness. Figure 7 depicts the results.

From Fig. 7, we find that critical and semi-critical links are railways that can be retrofitted. To elucidate further, the critical links include London–Folkestone, London–Bristol, London– Southampton, Bristol–Birmingham, and Southampton–Bristol. The semi-critical links are identified as Liverpool–Felixstowe and Felixstowe–London. Moving toward resilience, all critical and semi-critical links need to be retrofitted.

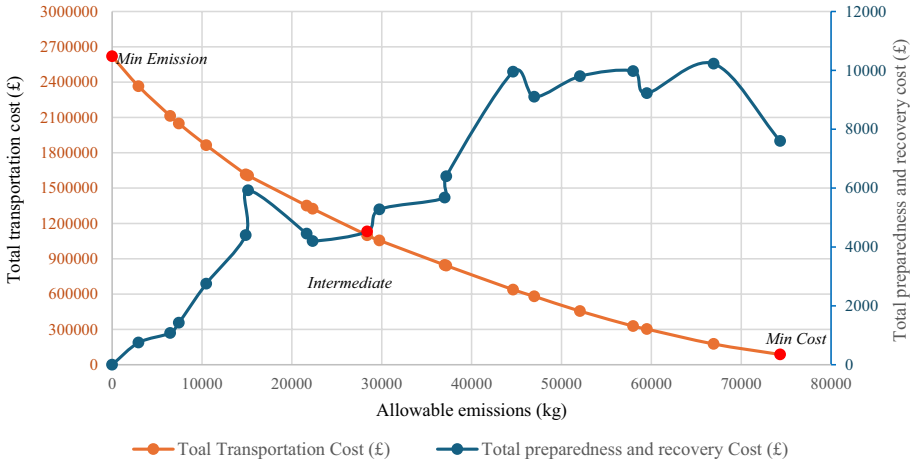


Fig. 6 The allowable emissions (kg) versus total transportation cost (£) and total preparedness and recovery cost (£)

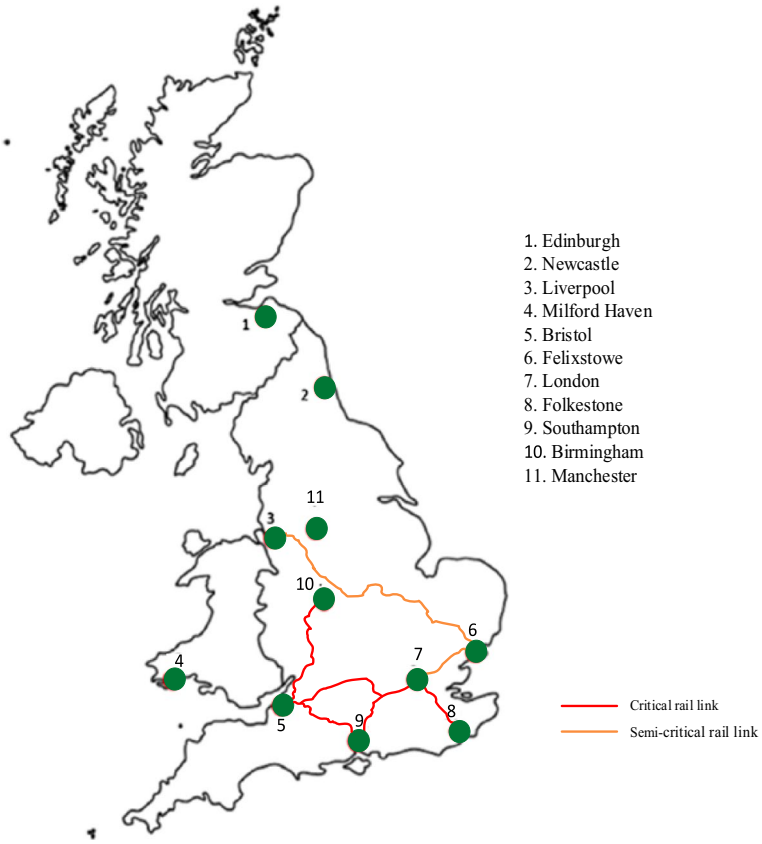


Fig. 7 The critical and semi-critical links to pre-disaster retrofitting

4.6 Value of the stochastic solution

To compare the performance of the proposed two-stage stochastic model with an expected value approach, two measures of value of the stochastic solution (*VSS*) and relative value of the stochastic solution ($\overline{VSS}$) are utilized. These two measures are used to compute the cost-saving that can be achieved by solving the two-stage stochastic model instead of its deterministic counterpart (see Birge (1982) for more details).

VSS and $\overline{VSS}$ are obtained using the following equations, respectively:

$$VSS = z^*_{det} - z^*_{stoc} \tag{32}$$

$$\overline{VSS} = \frac{z^*_{det} - z^*_{stoc}}{z^*_{det}} \tag{33}$$

where z^*_{stoc} is the optimal solution of the two-stage stochastic model and z^*_{det} is the objective value under the expected value approach. In other words, z^*_{det} is the optimal solution of the stochastic problem with first-stage variables fixed to the values of the optimal solution of the deterministic problem. The *VSS* clearly shows the benefit of adopting the stochastic

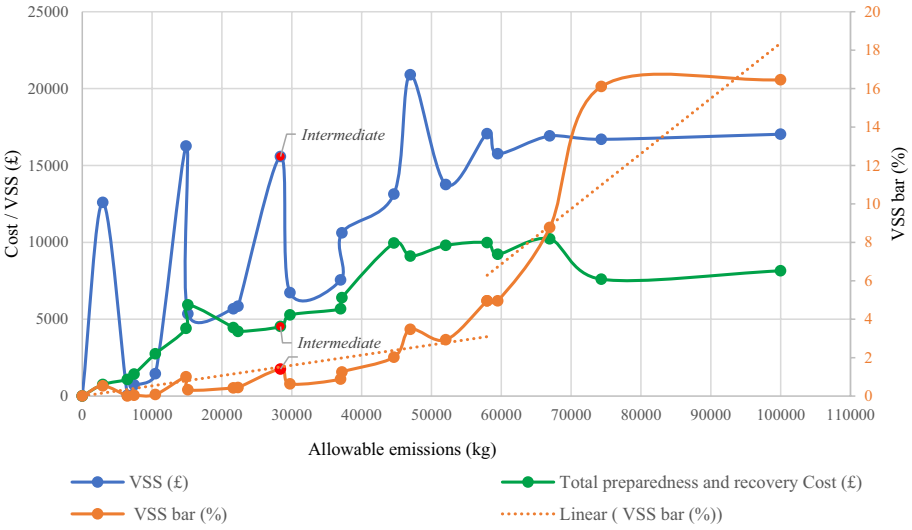


Fig. 8 The trend of allowable emissions (kg) versus VSS (£) and $\overline{VSS}$ (%)

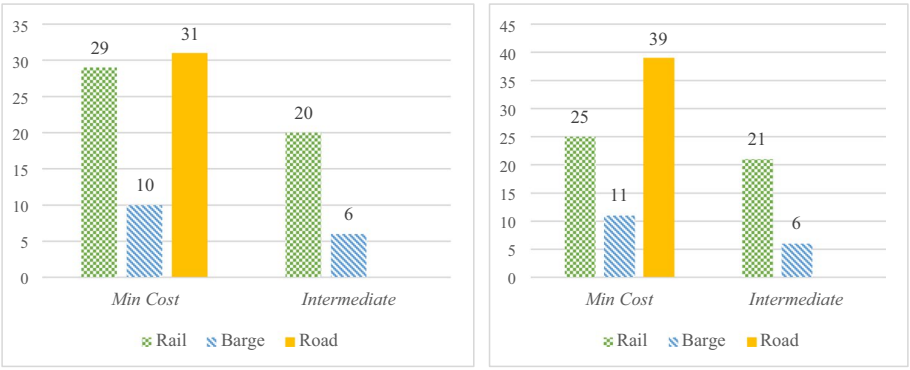
model instead of the deterministic formulation. The high values of *VSS* demonstrate that there is a significant improvement from considering disruption uncertainty in the system. Figure 8 shows different VSS and $\overline{VSS}$, obtained from solving the problems with different ϵ values.

Figure 8 shows that both *VSS* and $\overline{VSS}$ measures have an increasing trend as ϵ value grows. It is recalled that lower values of ϵ lead to greener systems. The most cost-saving is about £20,900, corresponding to the allowable emissions of 46,982 kg and the satisfied demand rate of 80.1%. However, the $\overline{VSS}$ is 3.47% in this point. It is worth noting that decreasing ϵ value leads to a greener system. The results also illustrate that in a brown system (when ϵ value is set at its maximum), more than 16% cost saving would be achieved by taking pre-disaster or post-disaster actions.

When the system is green and allowable emissions value is low, $\overline{VSS}$ is minor, meaning that the cost-saving obtained from the stochastic formulation is negligible. Moving towards a brown system can improve the $\overline{VSS}$ to a maximum of 16.46%. The slope of this graph intensifies at the point of 59,503 kg of allowable emissions value, corresponding to $\overline{VSS}$ of 4.95%, and jumps to $\overline{VSS} = 8.77\%$, with an increasing 12.5% of allowable emissions value. These observations can suggest that the benefits of the resilience consideration will be more pronounced when the system is brown.

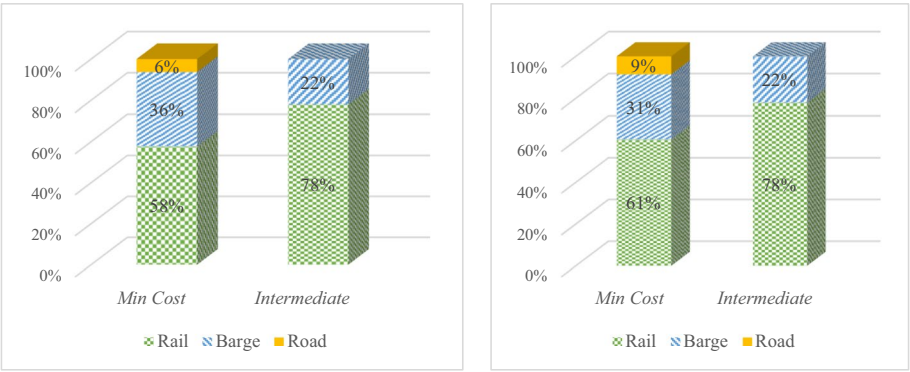
4.7 Impact of greenness on the transportation decisions

In this section, we investigate the impact of greenness on the optimal transportation decisions for the two following cases. In the first case (a), the fixed cost of trains is set to 100, while in the second case (b) the fixed cost of trains is set to 300. All other parameters are the same for both cases including fixed cost of trucks (equal to 100) and barges (equal to 300). A Pareto front is generated for each case, from which two points are selected to be



Case a. Case b.

Fig. 9 Total number of vehicles



Case a. Case b.

Fig.10 Percentage of demand assigned to each transportation mode

analyzed: *Min Cost* solution and *Intermediate* solution. As described in Sect. 5.2, *Min Cost* is the extreme point of the Pareto front, while *Intermediate* is the middle point of the Pareto front which incorporates greenness to the transportation planning.

Considering cases (a) and (b), Figs. 9 and 10 show the optimal number of vehicles and their share to deliver the orders, respectively.

According to Fig. 9, when only economic objective is regarded (i.e., *Min Cost* solution), a considerable number of trucks is used. While the number of trucks is more than the other vehicles in this situation, it does not mean that the road mode has the greatest share of delivery. As can be seen in Fig. 10, rail and barge modes deliver the highest percentage of orders. When emission concerns are incorporated (i.e., *Intermediate* solution), only barges and trains are utilized to deliver the orders. A comparison between cases (a) and (b) reveals that the number of vehicles varies with variation in train fixed cost. For the *Min Cost* solution, with an increase in the train fixed cost, the number of trains decreases from 29 to 25, and more trucks and barges are used to compensate. However, for the *Intermediate* solution, the percentage amount of demand assigned to transportation modes is robust to the variation in the train cost.

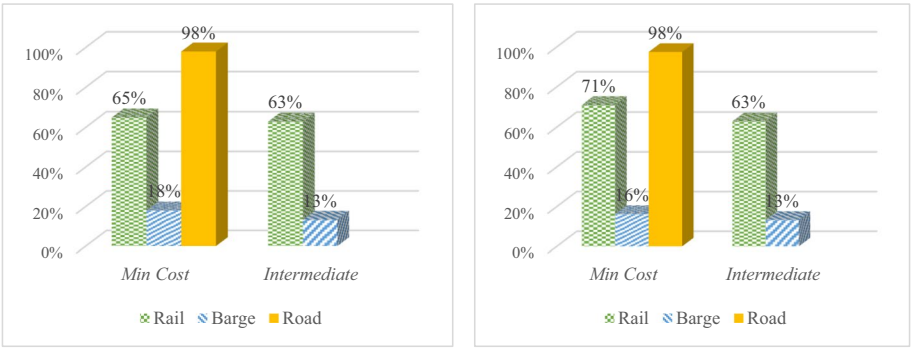


Fig. 11 Average capacity utilization (%)

Considering the *Min cost* and *Intermediate* solutions, Fig. 11 shows the average capacity utilization (%) of the transportation modes for the cases (a) and (b). For the *Min cost* solution, the capacity utilization of the road mode, as the least eco-friendly transportation mode in the problem, is about 98%. The least capacity utilization is related to the barge mode. In the *Intermediate* solution, the capacity utilization is decreased, compared to the *Min Cost* solution. This can be justified as follows. To cut down the emissions, fewer containers are utilized in the network. Comparing *Min Cost* solutions of the category (a) and (b) in Fig. 11, we find that the capacity utilization of the railroad mode improves from 65% to 71%. Another interesting observation is that the capacity utilization of the *Intermediate* solution stays unchanged with an increase in the fixed costs.

5 Discussion and managerial implications

The analysis of the numerical results presented in the previous section can provide valuable managerial insights and practical implications, guiding transport companies in their efforts to enhance sustainability and resilience. In comparison with earlier research in the literature (Jabbarzadeh et al., 2020; Miller-Hooks et al., 2012; Qu et al., 2014), we highlight some of these results as follows:

1. Greening the intermodal transport network in the face of disruption risks is not without its costs. However, the proposed approach empowers transportation companies to navigate desired trade-offs between overall costs and carbon emissions. A key insight for intermodal transport companies is the potential to significantly decrease greenhouse gas emissions with only a marginal increase in costs. Our analysis has uncovered that, under disruption risks, transitioning from a brown setting to a partially green setting (*Intermediate* point) resulted in a remarkable 62% reduction in emissions. Contrasting this emission reduction with the findings of Qu et al. (2014), which amounted to about 45%, leads to an insightful observation. In the research by Qu et al. (2014), disruption scenarios were disregarded. This suggests that the significance of greening initiatives becomes more pronounced when intermodal transport is more susceptible to disrup-

tions, offering a greater opportunity to curtail emissions for companies at a higher risk of disruptions.

2. On the other hand, we have found that under disruption risks, it is more challenging to meet the entire demand at a low emissions level. For example, with emissions of 57,974 kg CO_2e, equivalent to 78% of the maximum emissions in our case study, only 90% of the demand is met. To ensure full demand satisfaction, it is necessary to raise the allowable emissions threshold for the brown setting. If a transport company needs to be simultaneously green and able to satisfy demand during disruption scenarios, external options such as contracting with insurance companies and outsourcing seem indispensable.
3. Building resilience against random disruptions entails costs. However, based on the numerical results outlined in the preceding section, it is advisable for transport companies to invest in preparedness and recovery actions such as emergency response plans, redundancy assessments, infrastructure maintenance, technology advancements, and personnel training. While these initiatives may entail upfront costs, they have the potential to yield up to a 16% reduction in overall costs when considering future expected total costs. This finding aligns with the conclusions of Miller-Hooks et al. (2012), who demonstrated an 18% improvement in resilience through preparedness and recovery activities. It is noteworthy that, in their study, where emission levels were not considered, recovery actions were found to be more effective than preparedness actions. In contrast, our results indicate that at lower emission levels, preparedness actions can be more effective than recovery actions.
4. Numerical experiments have revealed that the optimal investment for preparedness and recovery actions is contingent upon the maximum levels of carbon emissions. The proposed methodology provides a valuable tool for transport companies seeking to make optimal decisions regarding resilience activities based on the desired level of environmental sustainability. A key finding from the computational experiment underscores the existence of critical links that warrant protection through preparedness actions, irrespective of emission levels. This observation aligns with prior literature (Jabbarzadeh et al., 2020). In the presented case study, the critical railway links—London–Folkestone, London–Bristol, London–Southampton, Bristol–Birmingham, and Southampton–Bristol—are retrofitted across all degrees of environmental sustainability. The identification of such critical links through the proposed methodology is strongly recommended for companies, particularly those requiring resilience investments before determining their environmental sustainability goals.
5. Our numerical experiments have consistently demonstrated that intermodal transport offers superior benefits in terms of emissions and cost compared to unimodal transport. While Qu et al. (2014) arrived at a similar conclusion, their analysis lacked a resilience examination. A key deviation from the findings of Qu et al. (2014) lies in the capacity utilization (%) of transport modes within optimal solutions. Computational results demonstrate that our proposed methodology generates capacity utilization more resistant to fluctuations in fixed costs. For instance, in a partially green setting (*Intermediate* point), the percentage of demand allocated to transportation modes and their corresponding capacity utilization remain robust even amidst variations in train costs. Additionally, in situations where the transport network is susceptible to disruptions, our optimal solutions indicate a higher reliance on road transport. This is in contrast to the findings of Qu et al. (2014), who did not account for disruption risk and showed fewer trucks utilized in optimal solutions. While a higher number of trucks may initially seem less environmentally sustainable, a closer examination reveals that their increased presence

does not necessarily translate to them carrying the largest share of flows in the network. Railways transport the highest percentage of container flows, while roadways contribute the least. The augmented use of trucks serves the purpose of meeting demand when critical transport links are unavailable during disruptive events. A practical recommendation for intermodal transport companies is to leverage road transport as a backup mode to fulfill demands in situations where more environmentally friendly modes are inaccessible due to disruptions.

Tables 10 and 11 provide a detailed comparison between Qu et al. (2014) and our study, in terms of capacity utilization and total number of vehicles from each mode, alongside with the percentage of demands assigned to different modes in both case a and case b.

As mentioned, in our model, the number of trucks and the percentage of demand assigned to roadways are higher than those in Qu et al. (2014). This indicates that, when considering the resilience of the system, road transport is promoted as a solution in disruptive conditions.

6 Conclusion

This study investigated how resilience strategies can be incorporated into the tactical planning of an intermodal rail-road-sea transportation network to cope with random disruptions. A bi-objective stochastic model was developed to account for both economic and environmental dimensions of sustainability under a budget limitation for preparedness and recovery actions. The proposed stochastic model, at the first stage, determines the optimal number of vehicles as well as the best set of preparedness activities for the links. These decisions stay tuned across all disruption scenarios. In the second stage, variables including the flow of shipments along each link for each order are determined for each disruption scenario. To generate non-dominated solutions for the bi-objective model, an augmented ϵ -constraint method was applied for a case study in the UK. The disruption scenarios were designed based on historical data, then numerical experiments were implemented and analyzed in two categories based on the various fixed cost of modes.

The model provided a range of solutions, offering a trade-off between emissions, costs, and the percentage of satisfied demand. Results demonstrated the potential for a significant reduction in greenhouse gas emissions with only a marginal increase in costs. A comparison of three settings—(i) a brown system, (ii) a partially green system, and (iii) a green system—revealed that increasing the total transportation budget from an economically efficient brown system to a partially green system could lead to a significant decrease of about 62% in emissions. However, under disruption risks, achieving a high customer service level at a low emissions level proved challenging. For instance, to meet only 90% of the demand, at least 78% of the emissions of a brown system needed to be emitted.

To measure the probable cost-saving gained by solving the stochastic model instead of its deterministic counterpart, two measures of VSS and $\overline{VSS}$ were used. Numerical experiments indicated that investing approximately 9% of the total transportation cost into preparedness and recovery initiatives could yield a cost saving of more than 16%, ensuring the satisfaction of 100% of demand in a brown setting. In a partially green setting, this cost saving was about 2%, requiring half of this budget for pre-disaster and post-disaster activities. Based on the result, the critical and semi-critical links were identified: these could

Table 10 Comparing the results with Qu et al. (2014) (case a)

	Fixed cost of trucks = 100 Fixed cost of trains = 100 Fixed cost of barges = 300								
	Truck			Rail			Barge		
	Capacity utilization (%)	Assigned percentage of demand	Total number	Capacity utilization (%)	Assigned percentage of demand	Total number	Capacity utilization (%)	Assigned percentage of demand	Total number
Qu et al. (2014)	NA	0%	0	79%	64%	31	21%	36%	9
The proposed study	98%	6%	31	65%	58%	29	18%	36%	10

Table 11 Comparing the results with Qu et al. (2014) (case b)

	Fixed cost of trucks = 100 Fixed cost of trains = 300 Fixed cost of barges = 300								
	Truck			Rail			Barge		
	Capacity utilization (%)	Assigned percentage of demand	Total number	Capacity utilization (%)	Assigned percentage of demand	Total number	Capacity utilization (%)	Assigned percentage of demand	Total number
Qu et al. (2014)	100%	1%	6	87%	55%	25	23%	44%	10
The proposed study	98%	9%	39	71%	61%	25	16%	31%	11

be retrofitted by implementing preparedness activities. The analysis further revealed that, despite a road mode capacity utilization of 98%, it served less than 9% of demand due to higher emissions, while railways transported the greatest percentage of container flows.

Future research could examine other social and environmental sustainability measures such as noise pollution, accidents, and congestion under different resource constraints. The proposed optimization approach can also be extended to determine other operational decisions, including human resource and vehicle scheduling decisions.

Acknowledgements This research was supported by grants from The Natural Sciences and Engineering Research Council of Canada (NSERC), Fonds de Recherche du Québec – Nature et technologies (FRQNT), and Centre interdisciplinaire de recherche en opérationnalisation du développement durable (CIRODD).

Funding Natural Sciences and Engineering Research Council of Canada, Fonds de recherche du Québec – Nature et technologies, CIRODD

Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no conflicts of interest to disclose.

References

Accenture report. (2021). *State of supply chains, In the eye of the storm*. Retrieved from https://www.accenture.com/ca-en/insights/consulting/coronavirus-supply-chain-disruption

Ahmadi-Javid, A., & Seddighi, A. H. (2013). A location-routing problem with disruption risk. *Transportation Research Part E: Logistics and Transportation Review, 53*, 63–82. https://doi.org/10.1016/j.tre.2013.02.002

Akyüz, M. H., Dekker, R., & Sharif Azadeh, S. (2023). Partial and complete replanning of an intermodal logistic system under disruptions. *Transportation Research Part e: Logistics and Transportation Review, 169*, 102968. https://doi.org/10.1016/j.tre.2022.102968

Allehashemi, T., Amin, S. H., & Zolfaghari, S. (2022). A proposed multi-objective model for cellphone closed-loop supply chain optimization based on fuzzy QFD. *Expert Systems with Applications, 210*, 118577. https://doi.org/10.1016/j.eswa.2022.118577

Archetti, C., Peirano, L., & Speranza, M. G. (2022). Optimization in multimodal freight transportation problems: A Survey. *European Journal of Operational Research, 299*(1), 1–20. https://doi.org/10.1016/j.ejor.2021.07.031

Azad, N., Hassini, E., & Verma, M. (2016). Disruption risk management in railroad networks: An optimization-based methodology and a case study. *Transportation Research Part B: Methodological, 85*, 70–88. https://doi.org/10.1016/j.trb.2016.01.001

Balaman, ŞY., Matopoulos, A., Wright, D. G., & Scott, J. (2018). Integrated optimization of sustainable supply chains and transportation networks for multi technology bio-based production: A decision support system based on fuzzy ε-constraint method. *Journal of Cleaner Production, 172*, 2594–2617. https://doi.org/10.1016/j.jclepro.2017.11.150

Baykasoğlu, A., Subulan, K., Taşan, A. S., & Dudaklı, N. (2018). A review of fleet planning problems in single and multimodal transportation systems. *Transportmetrica a: Transport Science, 15*(2), 631–697. https://doi.org/10.1080/23249935.2018.1523249

BBC News. (2021). *Russon, M-A, "The cost of the Suez Canal blockage". BBC News. 29 March 2021. Archived from the original on 29 March 2021*. Retrieved from https://www.bbc.com/news/business-56559073

Benali, N., & Feki, R. (2018). Evaluation of the relationship between freight transport, energy consumption, economic growth and greenhouse gas emissions: The VECM approach. *Environment, Development and Sustainability, 22*(2), 1039–1049. https://doi.org/10.1007/s10668-018-0232-x

Bérubé, J.-F., Gendreau, M., & Potvin, J.-Y. (2009). An exact -constraint method for bi-objective combinatorial optimization problems: Application to the traveling salesman problem with profits. *European Journal of Operational Research, 194*(1), 39–50. https://doi.org/10.1016/j.ejor.2007.12.014

Birge, J. R. (1982). The value of the stochastic solution in stochastic linear programs with fixed recourse. *Mathematical Programming, 24*(1), 314–325. https://doi.org/10.1007/bf01585113

Budiyanto, M. A., Huzaifi, M. H., & Sirait, S. J. (2019). Estimating CO_2 emissions in a container port based on modality movement in the terminal area. *International Journal of Technology, 10*(8), 1618. https://doi.org/10.14716/ijtech.v10i8.3508

Cabinet Office. (2017). *National Risk Register Of Civil Emergencies*. Retrieved from www.official-documents.gov.uk

Caris, A., Macharis, C., & Janssens, G. K. (2008). Planning problems in intermodal freight transport: Accomplishments and prospects. *Transportation Planning and Technology, 31*(3), 277–302. https://doi.org/10.1080/03081060802086397

Chen, L., & Miller-Hooks, E. (2012). Resilience: An indicator of recovery capability in intermodal freight transport. *Transportation Science, 46*(1), 109–123. https://doi.org/10.1287/trsc.1110.0376

Chen, R., & Zhang, Y. (2023). Freight transport structure evaluation and optimization toward sustainable development: New evidence from the SBM-DEA model with undesirable outputs. *Environment, Development and Sustainability.* https://doi.org/10.1007/s10668-023-03192-2

Crainic, T. G., & Kim, K. H. (2007). Intermodal transportation, book chapter. *Handbook in OR & MS, 14*, 467–537. https://doi.org/10.1016/S0927-0507(06)14008-6

Demir, E., Burgholzer, W., Hrušovský, M., Arıkan, E., Jammernegg, W., & Woensel, T. V. (2016). A green intermodal service network design problem with travel time uncertainty. *Transportation Research Part B: Methodological, 93*, 789–807. https://doi.org/10.1016/j.trb.2015.09.007

Demir, E., Hrušovský, M., Jammernegg, W., & Van Woensel, T. (2019). Green intermodal freight transportation: Bi-objective modelling and analysis. *International Journal of Production Research, 57*(19), 6162–6180. https://doi.org/10.1080/00207543.2019.1620363

Demir, E., Huang, Y., Scholts, S., & Van Woensel, T. (2015). A selected review on the negative externalities of the freight transportation: Modeling and pricing. *Transportation Research Part E: Logistics and Transportation Review, 77*, 95–114. https://doi.org/10.1016/j.tre.2015.02.020

Eurostat. (2023). *Freight transport statistics - modal split*. Retrieved from https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Freight_transport_statistics_-_modal_split#Modal_split_of_freight_transport_in_the_EU

Fallahi, N., Hafezalkotob, A., Raissi, S., & Ghezavati, V. (2023). A game theoretic approach to sustainable freight transportation: Competition between green, non-green and semi-green transportation networks under government sustainable policies. *Environment, Development and Sustainability.* https://doi.org/10.1007/s10668-023-03115-1

Fotuhi, F., & Huynh, N. (2016). Reliable intermodal freight network expansion with demand uncertainties and network disruptions. *Networks and Spatial Economics, 17*(2), 405–433. https://doi.org/10.1007/s11067-016-9331-0

Fotuhi, F., & Huynh, N. (2018). A reliable multi-period intermodal freight network expansion problem. *Computers & Industrial Engineering, 115*, 138–150. https://doi.org/10.1016/j.cie.2017.11.007

Ghaderi, A., & Burdett, R. L. (2019). An integrated location and routing approach for transporting hazardous materials in a bi-modal transportation network. *Transportation Research Part E: Logistics and Transportation Review, 127*, 49–65. https://doi.org/10.1016/j.tre.2019.04.011

The Guardian. (2021). *Safi, M, & Farrer, M., "Suez canal: Ever Given ship partially refloated but bow still stuck". The Guardian. Archived from the original on 29 March 2021*. Retrieved from https://www.theguardian.com/world/2021/mar/29/suez-canal-attempt-re-float-ever-given-delay-salvage-tugboats

Hamrouni, B., Bourouis, A., Korichi, A., & Brahmi, M. (2021). Explainable ontology-based intelligent decision support system for business model design and sustainability. *Sustainability, 13*(17), 9819. https://doi.org/10.3390/su13179819

Hartillo-Hermoso, M. I., Jiménez-Tafur, H., & Ucha-Enríquez, J. M. (2020). An exact algebraic ϵ-constraint method for bi-objective linear integer programming based on test sets. *European Journal of Operational Research, 282*(2), 453–463. https://doi.org/10.1016/j.ejor.2019.09.032

Hasani Goodarzi, A., Zegordi, S. H., Alpan, G., Nakhai Kamalabadi, I., & Husseinzadeh Kashan, A. (2020). Reliable cross-docking location problem under the risk of disruptions. *Operational Research*. https://doi.org/10.1007/s12351-020-00583-5

He, Y., Wu, T., Zhang, C., & Liang, Z. (2015). An improved MIP heuristic for the intermodal hub location problem. *Omega, 57*, 203–211.

He, Z., Navneet, K., van Dam, W., & Van Mieghem, P. (2021). Robustness assessment of multimodal freight transport networks. *Reliability Engineering & System Safety, 207*, 107315. https://doi.org/10.1016/j.ress.2020.107315

Hrušovský, M., Demir, E., Jammernegg, W., & Van Woensel, T. (2016). Hybrid simulation and optimization approach for green intermodal transportation problem with travel time uncertainty. *Flexible Services and Manufacturing Journal, 30*(3), 486–516. https://doi.org/10.1007/s10696-016-9267-1

Hrušovský, M., Demir, E., Jammernegg, W., & Van Woensel, T. (2021). Real-time disruption management approach for intermodal freight transportation. *Journal of Cleaner Production, 280*, 124826. https://doi.org/10.1016/j.jclepro.2020.124826

Hussein, M., Karam, A., Eltoukhy, A. E. E., Darko, A., & Zayed, T. (2023). Optimized multimodal logistics planning of modular integrated construction using hybrid multi-agent and metamodeling. *Automation in Construction, 145*, 104637. https://doi.org/10.1016/j.autcon.2022.104637

IEA (2020). Energy Technology Perspectives 2020. Retrieved from https://www.iea.org/reports/energy-technology-perspectives-2020

IEA. (2021). *Global Energy Review: CO2 Emissions in 2020*. Retrieved from https://www.iea.org/articles/global-energy-review-co2-emissions-in-2020

IEA. (2023). *Greenhouse Gas Emissions from Energy*. Retrieved from https://www.iea.org/data-and-statistics/data-product/greenhouse-gas-emissions-from-energy

Ishfaq, R. (2012). Resilience through flexibility in transportation operations. *International Journal of Logistics Research and Applications, 15*(4), 215–229. https://doi.org/10.1080/13675567.2012.709835

Ishfaq, R. (2017). Intermodal shipments as recourse in logistics disruptions. *Journal of the Operational Research Society, 64*(2), 229–240. https://doi.org/10.1057/jors.2012.40

Jabbarzadeh, A., Azad, N., & Verma, M. (2020). An optimization approach to planning rail hazmat shipments in the presence of random disruptions. *Omega, 96*, 102078. https://doi.org/10.1016/j.omega.2019.06.004

Jackson, B., & Finney, I. (2016). *Evaluating UK storm frequency and ECMWF forecast accuracy of extreme wind events*. Retrieved from https://www.ecmwf.int/sites/default/files/elibrary/2016/16440-evaluating-uk-storm-frequency-and-ecmwf-forecast-accuracy-extreme-wind-events.pdf

Kabadurmus, O., & Erdogan, M. S. (2020). Sustainable, multimodal and reliable supply chain design. *Annals of Operations Research, 292*(1), 47–70. https://doi.org/10.1007/s10479-020-03654-0

Ke, G. Y. (2020). Managing rail-truck intermodal transportation for hazardous materials with random yard disruptions. *Annals of Operations Research*. https://doi.org/10.1007/s10479-020-03699-1

Ke, G. Y., & Verma, M. (2021). A framework to managing disruption risk in rail-truck intermodal transportation networks. *Transportation Research Part e: Logistics and Transportation Review, 153*, 102340. https://doi.org/10.1016/j.tre.2021.102340

Kuzmicz, K. A., & Pesch, E. (2019). Approaches to empty container repositioning problems in the context of Eurasian intermodal transportation. *Omega, 85*, 194–213.

Lee, S.-K., Yoo, S., & Cheong, T. (2017). Sustainable EOQ under lead-time uncertainty and multimodal transport. *Sustainability, 9*(3), 476. https://doi.org/10.3390/su9030476

Li, J., Jing, K., Khimich, M., & Shen, L. (2023). Optimization of green containerized grain supply chain transportation problem in Ukraine considering disruption scenarios. *Sustainability, 15*(9), 7620. https://doi.org/10.3390/su15097620

Liotta, G., Stecca, G., & Kaihara, T. (2015). Optimisation of freight flows and sourcing in sustainable production and transportation networks. *International Journal of Production Economics, 164*, 351–365. https://doi.org/10.1016/j.ijpe.2014.12.016

Loveproperty, & S., Coffey. (2022). Top 10 places most likely to flood in the UK. Retrieved from https://www.loveproperty.com/news/83076/top-10-places-most-likely-to-flood-in-britain#:~:text=Hull%20has%20been%20named%20as,average%20of%200.7%20per%20cent

Lu, Y., Lang, M., Yu, X., & Li, S. (2019). A sustainable multimodal transport system: The two-echelon location-routing problem with consolidation in the Euro-China expressway. *Sustainability, 11*(19), 5486. https://doi.org/10.3390/su11195486

Maiyar, L. M., & Thakkar, J. J. (2019a). Environmentally conscious logistics planning for food grain industry considering wastages employing multi objective hybrid particle swarm optimization. *Transportation Research Part E: Logistics and Transportation Review, 127*, 220–248. https://doi.org/10.1016/j.tre.2019.05.006

Maiyar, L. M., & Thakkar, J. J. (2019b). Modelling and analysis of intermodal food grain transportation under hub disruption towards sustainability. *International Journal of Production Economics, 217*, 281–297. https://doi.org/10.1016/j.ijpe.2018.07.021

Maiyar, L. M., & Thakkar, J. J. (2019c). Robust optimisation of sustainable food grain transportation with uncertain supply and intentional disruptions. *International Journal of Production Research, 58*(18), 5651–5675. https://doi.org/10.1080/00207543.2019.1656836

Mavrotas, G. (2009). Effective implementation of the ε-constraint method in multi-objective mathematical programming problems. *Applied Mathematics and Computation, 213*(2), 455–465. https://doi.org/10.1016/j.amc.2009.03.037

Mavrotas, G., & Florios, K. (2013). An improved version of the augmented ε-constraint method (AUGMECON2) for finding the exact pareto set in multi-objective integer programming problems. *Applied Mathematics and Computation, 219*(18), 9652–9669. https://doi.org/10.1016/j.amc.2013.03.002

Miller-Hooks, E., Zhang, X., & Faturechi, R. (2012). Measuring and maximizing resilience of freight transportation networks. *Computers & Operations Research, 39*(7), 1633–1643. https://doi.org/10.1016/j.cor.2011.09.017

Monios, J. (2016). *Institutional challenges to intermodal transport and logistics*

Monios, J. (2015). Integrating intermodal transport with logistics: A case study of the UK retail sector. *Transportation Planning and Technology, 38*(3), 347–374. https://doi.org/10.1080/03081060.2015.1008798

Mostert, M., Caris, A., & Limbourg, S. (2017). Intermodal network design: A three-mode bi-objective model applied to the case of Belgium. *Flexible Services and Manufacturing Journal, 30*(3), 397–420. https://doi.org/10.1007/s10696-016-9275-1

Nocera, F., Contento, A., & Gardoni, P. (2024). Risk analysis of supply chains: The role of supporting structures and infrastructure. *Reliability Engineering & System Safety, 241*, 109623. https://doi.org/10.1016/j.ress.2023.109623

Okyere, S., Yang, J. Q., Aning, K. S., & Zhan, B. (2019). Review of sustainable multimodal freight transportation system in African developing countries: evidence from Ghana. *International Journal of Engineering Research in Africa, 41*, 155–174. https://doi.org/10.4028/www.scientific.net/JERA.41.155

Peng, Z., & Wu, Q. (2019). Evaluation of the relationship between energy consumption, economic growth, and CO_2 emissions in China' transport sector: The FMOLS and VECM approaches. *Environment, Development and Sustainability, 22*(7), 6537–6561. https://doi.org/10.1007/s10668-019-00498-y

Pudasaini, P. (2021). Integrated planning of downstream petroleum supply chain: a multi-objective stochastic approach. *Operations Research Perspectives, 8*, 100189. https://doi.org/10.1016/j.orp.2021.100189

Qi, Y., Harrod, S., Psaraftis, H. N., & Lang, M. (2022). Transport service selection and routing with carbon emissions and inventory costs consideration in the context of the Belt and Road Initiative. *Transportation Research Part e: Logistics and Transportation Review, 159*, 102630. https://doi.org/10.1016/j.tre.2022.102630

Qu, Y., Bektaş, T., & Bennell, J. (2014). Sustainability SI: Multimode multicommodity network design model for intermodal freight transportation with transfer and emission costs. *Networks and Spatial Economics, 16*(1), 303–329. https://doi.org/10.1007/s11067-014-9227-9

Resat, H. G., & Turkay, M. (2019). A bi-objective model for design and analysis of sustainable intermodal transportation systems: A case study of Turkey. *International Journal of Production Research, 57*(19), 6146–6161. https://doi.org/10.1080/00207543.2019.1587187

Rosyida, E. E., Santosa, B., Pujawan, I. N., & Guo, J. (2020). Freight route planning in intermodal transportation network to deal with combinational disruptions. *Cogent Engineering, 7*(1), 1805156. https://doi.org/10.1080/23311916.2020.1805156

Sarhadi, H., Tulett, D. M., & Verma, M. (2017). An analytical approach to the protection planning of a rail intermodal terminal network. *European Journal of Operational Research, 257*(2), 511–525. https://doi.org/10.1016/j.ejor.2016.07.036

Sikder, M., Wang, C., Yeboah, F. K., & Wood, J. (2022). Driving factors of CO2 emission reduction in the logistics industry: an assessment of the RCEP and SAARC economies. *Environment, Development and Sustainability*, 1–31

SteadieSeifi, M., Dellaert, N. P., Nuijten, W., Woensel, T. V., & Raoufi, R. (2014). Multimodal freight transportation planning: A literature review. *European Journal of Operational Research, 233*, 1–15.

Sun, Y., Hrušovský, M., Zhang, C., & Lang, M. (2018). A time-dependent Fuzzy programming approach for the green multimodal routing problem with rail service capacity uncertainty and road traffic congestion. *Complexity, 2018*, 1–22. https://doi.org/10.1155/2018/8645793

Sun, Y., Li, X., Liang, X., & Zhang, C. (2019). A bi-objective Fuzzy credibilistic chance-constrained programming approach for the hazardous materials road-rail multimodal routing problem under uncertainty and sustainability. *Sustainability, 11*(9), 2577. https://doi.org/10.3390/su11092577

Sun, Y., Yu, N., & Huang, B. (2021). Green road–rail intermodal routing problem with improved pickup and delivery services integrating truck departure time planning under uncertainty: An interactive fuzzy programming approach. *Complex & Intelligent Systems, 8*(2), 1459–1486. https://doi.org/10.1007/s40747-021-00598-1

Tamannaei, M., Zarei, H., & Rasti-Barzoki, M. (2021). A game theoretic approach to sustainable freight transportation: Competition between road and intermodal road–rail systems with government intervention. *Transportation Research Part B: Methodological, 153*, 272–295. https://doi.org/10.1016/j.trb.2021.09.002

Uddin, M., & Huynh, N. (2016). Routing model for multicommodity freight in an intermodal network under disruptions. *Transportation Research Record: Journal of the Transportation Research Board, 2548*(1), 71–80. https://doi.org/10.3141/2548-09

Uddin, M., & Huynh, N. (2019). Reliable routing of road-rail intermodal freight under uncertainty. *Networks and Spatial Economics, 19*(3), 929–952. https://doi.org/10.1007/s11067-018-9438-6

United Kingdom Government. (2022). *Transport and Environment Statistics 2022*. Retrieved from https://www.gov.uk/government/statistics/transport-and-environment-statistics-2022/transport-and-environment-statistics-2022#:~:text=transport%20is%20the%20largest%20emitting,each%20local%20authority%20in%202020

Wagenaar, J., Fragkos, I., & Zuidwijk, R. (2021). Integrated planning for multimodal networks with disruptions and customer service requirements. *Transportation Science, 55*(1), 196–221. https://doi.org/10.1287/trsc.2020.1006

Wang, C.-N., Nhieu, N.-L., Chung, Y.-C., & Pham, H.-T. (2021). Multi-objective optimization models for sustainable perishable intermodal multi-product networks with delivery time window. *Mathematics, 9*(4), 379. https://doi.org/10.3390/math9040379

Williams, H. P. (2013). *Model building in mathematical programming* (5th ed.): John Wiley & Sons Ltd

Yang, Z., Xin, X., Chen, K., & Yang, A. (2021). Coastal container multimodal transportation system shipping network design—toll policy joint optimization model. *Journal of Cleaner Production, 279*, 123340. https://doi.org/10.1016/j.jclepro.2020.123340

Zhao, Y., Xue, Q., Cao, Z., & Zhang, X. (2018). A two-stage chance constrained approach with application to stochastic intermodal service network design problems. *Journal of Advanced Transportation, 2018*, 1–18. https://doi.org/10.1155/2018/6051029

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.